



Tone, stress, quantity, and quality: prosodic patterns and tonal wug-tests in Žiri Slovenian

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ABSTRACT

In this paper, we report on the prosodic system of Žiri Slovenian, which displays an intricate set of interactions between tone, stress, quantity, and vowel quality. The prosodic restrictions are complex, exceeding the patterns observed in most other pitch-accented languages. Some of the patterns are well-motivated (e.g. the preference of High tone syllables to be stressed; the requirement for long vowels to have High tone) while others lack phonetic or phonological motivation (e.g. the requirement of long vowels to be footed). Žiri also displays a rare case of interaction of prosody with vowel quality: long [ɛɛ] must always be stressed and prefers the High tone on the second mora, which is not the case for any other vowel. To confirm the productivity of the observed patterns, we conduct a tonal wug experiment that tests the dependence of stress on vowel quality, word length, and tone. This paper brings forth a new instance of a phonetically unmotivated phonological process at the suprasegmental level, which appears to be less discussed than at the segmental level. We also discuss methodological issues arising from artificial experiments on tonal processes.

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1. Introduction

Many prosodic systems incorporate both stress and tone (Lehiste, 1970; Hyman, 2006). Ma'ya, for instance, has a tonal contrast on the final syllable as well as lexically contrastive stress (Remijsen, 2002). What is more common, however, is that stress and tone interact with one another. In Ayutla Mixtec, the position of stress is predictable from the distribution of tone, generally targeting the highest tone (de Lacy, 2002). In Choguita Rarámuri, stress is contrastive, and the stressed syllables contrast three tonal patterns; there are no tonal contrasts in unstressed syllables (Caballero & Carroll, 2015). A more complex situation is found in Bosnian-Croatian-Montenegrin-Serbian (BCMS; previously known as Serbo-Croatian in the literature), where stress is assigned to the syllable preceding a High tone, which targets ends of morphological domains, or else the word-initial syllable (Zec, 1999).

In this paper, we report novel data from the Žiri dialect of Slovenian, which also displays interactions between stress

and tone. Stress falls on the leftmost High tone, forming a trochaic foot (e.g. /pomórántʃp/ → [pɔ('móráán)tʃp] 'orange'). Conversely, tone is prosodically restricted in multiple ways. First, each Prosodic Word must have a High tone (e.g. /plénitʃp/ → [plɛ('níʃp)] 'diaper'). Second, each word has at most two High tones, and they must fall on adjacent syllables, that is, within the head foot. While [tɛɛ('vízi)] 'television' is attested, forms like *[tɛɛ('vízi)]jɔ are impossible because they have three High tones, and forms like *[tɛɛ('vízi)]jɔ are impossible because their High tones are not in adjacent syllables, regardless of how we foot these forms. Third, long vowels must have a High tone, which means that long vowels must be footed. The High tone can fall on the first mora of the long vowel, resulting in a falling tone (e.g. /bótáan/ → ['bótáan]) 'sick'; /kówáatʃ/ → [kɔ('wáatʃ)] 'blacksmith'), or on the second mora of the long vowel, resulting in a rising tone (e.g. /otráats/ → [p('tráats)] 'children'; /kórwaáw/ → ['kórwaáw]) 'bloody'). Fourth, in words with only short vowels, the trochee is aligned with the end of the word; the penultimate syllable receives High tone and stress (as seen above in /plénitʃp/ → [plɛ('níʃp)] 'diaper'). Thus, short vowels always have a High tone when stressed (so *[plɛ('níʃp)] is not possible), and can

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never have a High tone when they are unstressed (which rules out unattested *[plɛ(ˈnítsó)] or *[plɛ(ˈnítsɔ)]).

The complexity of the Žiri prosodic system, which requires reference to three suprasegmental properties, has been noted in the Slavic literature (Stanonik, 1977; Greenberg, 1987; Greenberg, 2003; Beguš, 2011). In the context of the Western South Slavic pitch accent, Žiri is considered to have an exceptionally large number of prosodic contrasts. In disyllables, for instance, BCMS contrasts a falling and rising pitch accent, but stress is always initial. Standard Slovenian distinguishes falling and rising pitch accents, but stress can fall on either syllable, resulting in four different combinations. In Žiri, there are seven patterns in total, five of which have initial stress. The larger number is because Žiri permits two non-adjacent pitch peaks, while the other Western South Slavic varieties allow only one. As far as we know, the only other Slavic varieties that approach Žiri are a few BCMS dialects (Kapović, 2008; Kapović, 2015:685ff.). The large number of prosodic combinations also distinguishes Žiri from other Indo-European pitch accent systems, such as in West (Gussenhoven, 2004; Köhnlein, 2016) and North Germanic (Riad, 2012; Myrberg & Riad, 2015), where disyllabic words typically have two distinctive lexical pitch accents.¹

The final interaction in Žiri concerns the interplay of prosodic and segmental properties. While all reported patterns thus far are true regardless of vowel quality, there is one exception. The long [ɛɛ] must always be stressed. For example, /zɔklɛ́nt/ → [zɔ(ˈklɛ́nt)] ‘to lock’ is attested, while words like *[ˈzɔklɛ́nt] are impossible. Other long vowels impose no restriction on stress (e.g. /pɔ́dtoók/ → [ˈpɔ́dtoók] ‘bolt’). Moreover, [ɛɛ] has rising pitch much more commonly than falling, especially in non-final syllables.

The interactions between segmental and suprasegmental features have generated a great deal of interest in phonological theory even though they are not all equally well-studied. The interaction between obstruent voicing and tone, for instance, is fairly well-understood: voiceless obstruents raise pitch and voiced obstruents lower pitch, and this phonetic effect can be phonologized (Hombert, 1978; Bradshaw, 1997; Bradshaw, 1999; Kingston, 2007; Donnelly, 2007; Lee, 2008; Pearce, 2007; Pearce, 2009; Tang, 2008; Coetzee, Beddor, Shedden, Styler, & Wissing, 2018; Gao & Kirby, 2024). For instance, in Siswati, a High tone becomes rising after a voiced obstruent (Odden, 2011).

Vowel height can also affect tone: typically, vowel height correlates with pitch (Ladd & Silverman, 1984; Ohala & Eukel, 1987; Whalen & Levitt, 1995; Fowler & Brown, 1997; Connell, 2002). This intrinsic fundamental frequency effect can be phonologized. In Awad Bing, for example, vowel-initial words have a Low tone, except for initial [i], which has a High tone (Cahill, 2001). An extensive cross-linguistic survey conducted by Becker and Jurgec (2017) reveals that the negative correlation between vowel height and tone is also attested: Ngizim has a predictable High tone on the low vowel

[a], but other vowels show no restriction (Schuh, 1971). Other vowel quality interactions are also attested: tense vowels select lower tones than lax vowels in Rengao (Gregerson, 1976). Beyond segmental features, sonority can also interact with tone (de Lacy, 2002). In Uspanteko, tone realization in disyllables depends on the sonority difference between the two syllables (Bennett & Henderson, 2013). Stress can also be involved. Interactions between vowel quality and stress have been reported for Standard Slovenian, where lax mid, but not other vowels avoid stress (Becker & Jurgec, 2020). In English, the final [i] pushes stress to the antipenultimate syllable, which is not observed with other vowels (Moore-Cantwell, 2020). Across languages, stress has a significant effect on vowel quality, both phonetically and phonologically, with unstressed syllables showing fewer vowel contrasts than stressed syllables (Fourakis, 1991; Pitermann, 2000; Crosswhite, 2001). Sonority can also affect stress assignment, with stress generally preferring high sonority vowels (Kenstowicz, 1997; de Lacy, 2006), even though some of these cases have been recently put into question (Rasin, 2017; Shih, 2016; Shih, 2018; Shih & de Lacy, 2020).

Žiri offers evidence that is not related to sonority: it is the markedness of [ɛɛ] that affects the position of stress and tone. This not only complements our understanding of possible interactions between segmental and suprasegmental properties, but also adds to the literature on phonetically unmotivated and unnatural processes. These have been discussed substantially in the literature (Blevins, 2004; Kiparsky, 2006; Kiparsky, 2008; Coetzee & Pretorius, 2010; Beguš, 2018; Beguš, 2019; Beguš, 2020; Beguš, 2022; Dąbkowski & Beguš, 2024; Beguš & Dąbkowski, 2025), but most of the discussion focuses on segmental patterns. Less work exists on phonetically unmotivated suprasegmental phonology. In this paper, we present a phonetically unmotivated process that targets the distribution of stress.

This paper consists of three parts. In Section 2, we describe the prosodic patterns in monosyllabic and disyllabic words. Our description of the Žiri pattern is based on the available literature (particularly Stanonik, 1977), complemented by original fieldwork elicitation of forms from native speakers. The transcriptions are corroborated by an acoustic analysis of pitch, intensity, and duration. In Section 3, we examine prosodic alternations. When words with initial High tone are preceded by a clitic (typically a preposition), a second High tone appears on the clitic with some words, resulting in stress retraction. We detail these patterns for the first time and complement them with acoustic analysis. Both sets of data reveal an unexpected interaction between vowel quality and prosody. In Section 4 we ask whether this interaction is an accidental gap or a productive component of Žiri grammar. In addition, we also ask whether there is a difference in terms of word length and pitch type. We conducted a perceptual wug-test and found that the participants’ judgments confirm the generalizations observed in the fieldwork data. Since nonce word experiments involving tone are particularly rare, the present study contributes to the body of literature on tonal wug-tests in various Chinese languages (Zhang, Lai, & Turnbull-Sailor, 2006; Zhang & Lai, 2010; Zhang & Liu, 2016). To our knowledge, the experiment conducted in this study is the first tonal wug-test that looks at pitch accent in particular.

¹ Beyond pitch accent, most Slavic languages have secondary stress, which remains beyond the scope of this paper (Gouskova & Roon, 2009; Gouskova, 2010; Gouskova & Becker, 2013; Jurgec, 2010; Newlin-Lukowicz, 2012; Łukaszewicz, 2018; Łukaszewicz & Molzcanow, 2018; Ronelle, 2019; see Jurgec, 2023 for an overview across Slavic languages).

2. Prosodic patterns in Žiri

The variety of Slovenian described in this paper is that of Žiri [ʒiˈri], a town in Western Slovenia with a population of 5,047 (Statistical Office of the Republic of Slovenia, 2024). The number of speakers of the dialect can be approximated from the population size, but the precise number of native speakers is unknown.

Žiri can be characterized as a pitch-accented variety of Slovenian. This means that tone interacts with other prosodic properties (Zec, 1999; Hyman, 2006; van der Hulst, 2011). In the case of Žiri, tone is dependent on vowel length, vowel quality, and word size. Tone furthermore determines the position of stress. Žiri displays a unique combination of prosodic factors that are not found to the same extent in other varieties of Slovenian or the closely related BCMS. For instance, the Standard Slovenian pitch accent distinction is limited to two contrastive contours (Falling and Rising) limited to the stressed and post-tonic syllable. Long vowels are possible only in stressed syllables (see Greenberg, 2003 for an overview). In BCMS, the distribution of pitch is even more restricted (stressed syllables only, no contrast in monosyllables), while the quantity distinctions can appear in stressed and all posttonic syllables (Ivić, 1958; Lehiste, 1961; Lehiste & Ivić, 1963; Lehiste, 1970; Lehiste & Ivić, 1986; Zec, 1999; Zsiga & Zec, 2013; Zec & Zsiga, 2022). Like BCMS, Žiri allows unstressed long vowels, but these can also further contrast tone, thus resulting in a greater number of attested prosodic combinations.

In this section, we first outline the methods (Section 2.1). Next, we introduce the vowel inventory, which is dependent on vowel length (Section 2.2). Then we look at the vowel length and tonal contrasts in monosyllables (Section 2.3), followed by disyllables (Section 2.4). We postpone the discussion of alternations until Section 3.

2.1. Methods

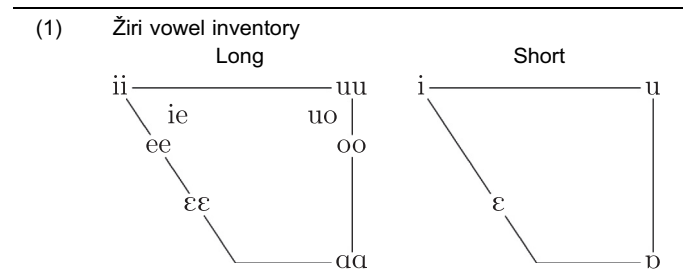
The data reported in Sections 2 and 3 is based on the material and analysis by Stanonik (1977) and Beguš (2011), as well as our original fieldwork, conducted in the Žiri Valley in the summer of 2018.

Participants consented to recording by signing informed consent forms approved by the Institutional Review Board (IRB) committee. A total of 12 participants (mean age = 68.7, median = 84, range from 23 to 92; the age of one participant is undisclosed) were recorded in elicitation sessions that lasted approximately 2 hours each. Recordings were conducted in participants' homes using a Sound Devices USB Pre 2 pre-amplifier (with 44.1 kHz sampling) and an AKG C544L cardioid condenser head-mounted microphone. Participants were shown pictures of objects chosen for elicitation from an English picture dictionary (Adelson-Goldstein & Shapiro, 2016) that contained no Slovenian words. They were asked to name objects and use them in phrases or short sentences in order to elicit responses with a preposition. The interviewer (one of the authors) instructed the participants in the Žiri dialect in order to avoid influences of Standard Slovenian.

A subset of the recordings was segmented in Praat (Boersma & Weenink, 1992). A full list of the forms is in the Appendix, Table 4. Statistical analysis was performed in R (Core Team, 2013).

2.2. Vowel inventory and vowel length

Žiri distinguishes long and short vowels, as in (1). There are eight distinctive long vowels and four short vowels (Stanonik, 1977). Notice also the rounding contrast in the low back vowels (long [ɑɑ] versus short [ɒ]) and the vowel height asymmetry between front and back vowels. The asymmetry plays a key role in the vowel quality restrictions, which apply only to the [ɛɛ]. Throughout this paper, we use doubled monophthongs to denote long vowels rather than the standard length symbol, following the practice in the analyses of BCMS (e.g. Inkelas & Zec, 1988; Zec, 1999). This representation allows us to more clearly represent the association of a tone to either the first or second mora of the long vowel, a crucial distinction that will be discussed in the following sections. The diphthongs are inherently long, which is also typical of other Slovenian dialects with quantity distinctions (Škofic & Kenda Jež, 2021).



Long and short vowels can appear in any position of the word (initial, medial, final), but there can only be one long vowel per word. Long vowels can be stressed or not, and the same is true for short vowels (Stanonik, 1977). Vowel length is thus not predictable, so we posit that the distinction between long and short vowels is underlying. This can be further supported by minimal pairs such as [vɛi] 'twist-PARTIC.M.PL' versus [vɛi] 'fairy-GEN.PL'. Contrastive vowel length makes Žiri similar to BCMS (Inkelas & Zec, 1988; Zec, 1999; Langston, 1997) and some varieties of Slovenian (Lenček, 1966; Bidwell, 1969; Toporišič, 1976/2000; Herry, 2000).

Phonetically, vowel length is realized as increased duration. Fig. 1 shows vowel duration by phonological length. Here we include monosyllabic data across all participants. The token counts for the data in this and all other figures are in Table 3 in the Appendix.

As we can see, long vowels are substantially longer (mean and median: 143 ms) than short vowels (mean: 88 ms; median: 89 ms). This differs from many other varieties of Slovenian where vowel length is no longer contrastive (see Jurgec, 2011; Jurgec, 2019 for further discussion).

2.3. Monosyllables

Next we look at the prosodic patterns in monosyllabic words. We first present the data, describe it, followed by a phonological analysis, which we then support with acoustic measurements. Monosyllabic words contain a single stressed vowel, which makes their prosody simpler than what is observed in disyllables. The vowels can differ in terms of quality, length and tone, as shown in (2).

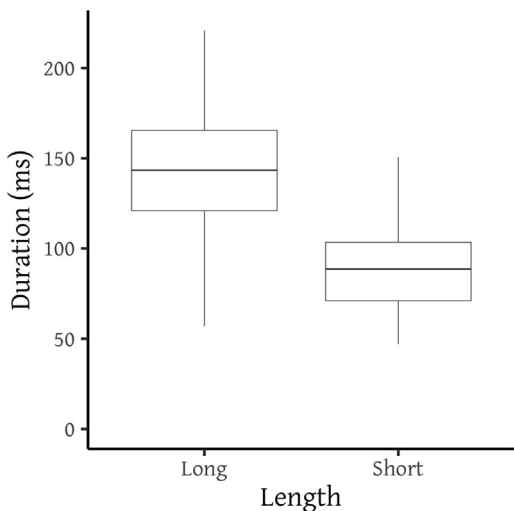


Fig. 1. Duration of vowels by phonological length in the elicited monosyllables. Phonologically long vowels include both falling (e.g. [méex] ‘bellows’) and rising contours ([xleépts] ‘loaf’), while the short vowels all bear a High tone ([túf] ‘light’). The box includes 50% of the data (interquartile range), with the middle line representing the median. The whiskers include the remaining values, as long as they are within 1.5 of the interquartile range. Outliers are represented as dots beyond that range; none are present in this dataset. These data only include monosyllables that are not preceded by a clitic; see Section 3 for further details.

(2) Distribution of tones in monosyllables (Stanonik, 1977)					
Long Falling		Long Rising		Short High	
zír	‘beechnut’	zír	‘Žiri’	zít	‘wall’
líet	‘ice’	spiét	‘bound’		
ýréex	‘sin’	deéts	‘male’		
tšéel	‘forehead’	één:	‘one’	zét	‘son-in-law’
drúujx	‘others’	uús	‘cart’	túf	‘light’
púot	‘path’	yuópts	‘muzzle’		
nóotf	‘night’	boóxx	‘God’		
ýráat	‘castle’	maájK	‘male’	króx	‘bread’

Among the words with long vowels, there are two distinct patterns. The first group of monosyllables have long vowels with a falling tone pattern. We transcribe these words with a High tone on the first mora of the long vowel, as in [úu]. The second pattern has a rising contour, which we transcribe with a High tone on the second mora of the long vowel, as in [uú]. In the Slavic literature, the falling pattern is termed the circumflex, while the rising is termed the acute (Gvozdanović, 1980; Bethin, 1998; Zec, 1999; Toporišič, 1976/2000; Greenberg, 2000; Langston, 2006). Finally, short vowels are pronounced with a High tone [ú].

Beyond the segmental inventory restrictions in (1), all combinations of vowel quality and tone are observed. It is worth pointing out, however, that Long Rising monosyllables are rarer than Long Falling (Stanonik, 1977:298). This has to do with the historical developments in Žiri, which are often determined by word length. Common Slovenian rising pitch developed into Long Falling in Žiri monosyllables (Beguš, 2011; Stanonik, 1977). For instance, while Žiri [ýréex] ‘sin’ and [nóotf] ‘night’ are Long Falling, the former had rising (acute) pitch in Common Slovenian while the latter retained its falling (circumflex) pitch. The Long Rising pitch observed in Žiri monosyllables is thus secondary and the result of segmental

deletions, analogy, and borrowing. Disyllables and longer words show other patterns. See Section 4.3 for further details. While these new Long Rising words are not rare by any measure, they do not quite reach the number of Long Falling words. For instance, Stanonik (1977) lists 94 Long Falling, but only 26 Long Rising monosyllables (or 22% of all long monosyllables). Žiri is not alone in this tonal asymmetry: in Standard Slovenian rising tone is less common than falling (37% versus 63% according to Jurgec, 2004).

To analyze these contrasts we make use of autosegmental representations (Goldsmith, 1976 et seq.). We propose that only High tone is phonologically active (3), as in BCMS. For the falling contours, the High tone is associated with the first mora, whereas for the rising contours, the High tone is linked to the second mora. Short vowels are monomoraic and have a single High tone.

(3) Tonal contrasts in monosyllables			
Long Falling		Long Rising	Short
H		H	H
lees		deets	zet

Monosyllables without a High tone are unattested. That is to say, each word must have at least one High tone. In the following sections, we will connect this gap with the interactions between stress and tone: stressed syllables must always have a High tone, and when the word has more than one, stress falls on the leftmost High tone.

The three-way contrast in monosyllables becomes clear when we examine the pitch tracks, which we present in Fig. 2. Pitch tracks were extracted from annotated files using a Praat script that measured pitch in 10 equidistant intervals for each vowel. The data was normalized by participant using R’s `normalize` function which gives the data in z-scores. The token counts for the data in this and all other figures are in Table 3 in the Appendix.²

The Long Falling pitch has a peak at the beginning, whereas the Long Rising pitch starts low and peaks at the end. Finally, the Short vowel has a relatively flat high pitch, that dips slightly towards the end. The vowel is also significantly shorter, but this is not represented in Fig. 2, as all vowels regardless of length have been split into ten intervals.

Even though Fig. 2 shows overall higher pitch for short vowels, this is not shared among all participants. We present individual data for six participants that had at least 4 tokens for each prosodic combination. Fig. 3. Participant 10, for instance, has a short vowel realization with the pitch maximum no higher than Long Falling pitch. We interpret the overall higher pitch in short vowels when compared to long vowels and the dip towards the end of the vowel as phonetic rather than phonolog-

² Rather than presenting pitch tracks of individual tokens, which is the practice in many studies, we analyze multiple pitch tracks from several speakers together. This approach yields a more reliable analysis of tonal trajectories in a language. In the figures provided, both previous and subsequent, we have opted for a descriptive statistical analysis over an inferential one. This is because a detailed phonetic realization of the contrasts, necessitating a very large annotated data set, is not the focus of this paper and is left for future work.

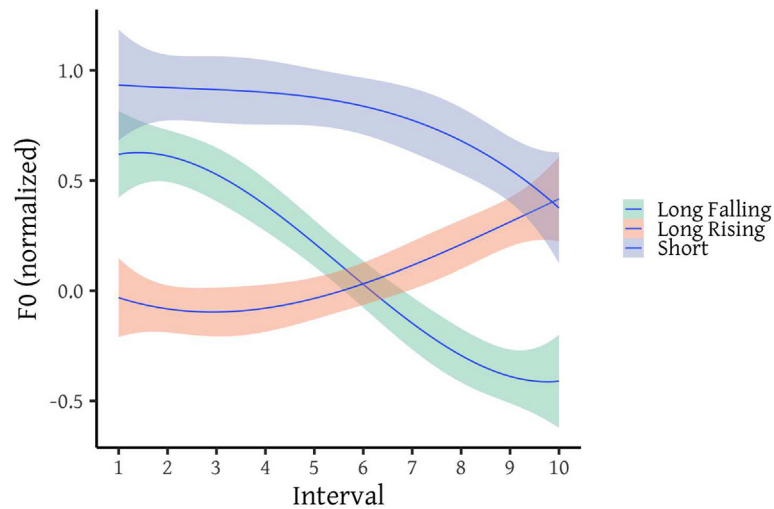


Fig. 2. Normalized pitch tracks in the elicited monosyllables. The Long Falling monosyllables include [méex] 'bellows', Long Rising [xleépts] 'loaf', and Short [túf] 'light' (2). The middle line for each group presents the smoothed trend line using the `lm` function with quadratic splines, and the shaded areas present 95% confidence intervals.

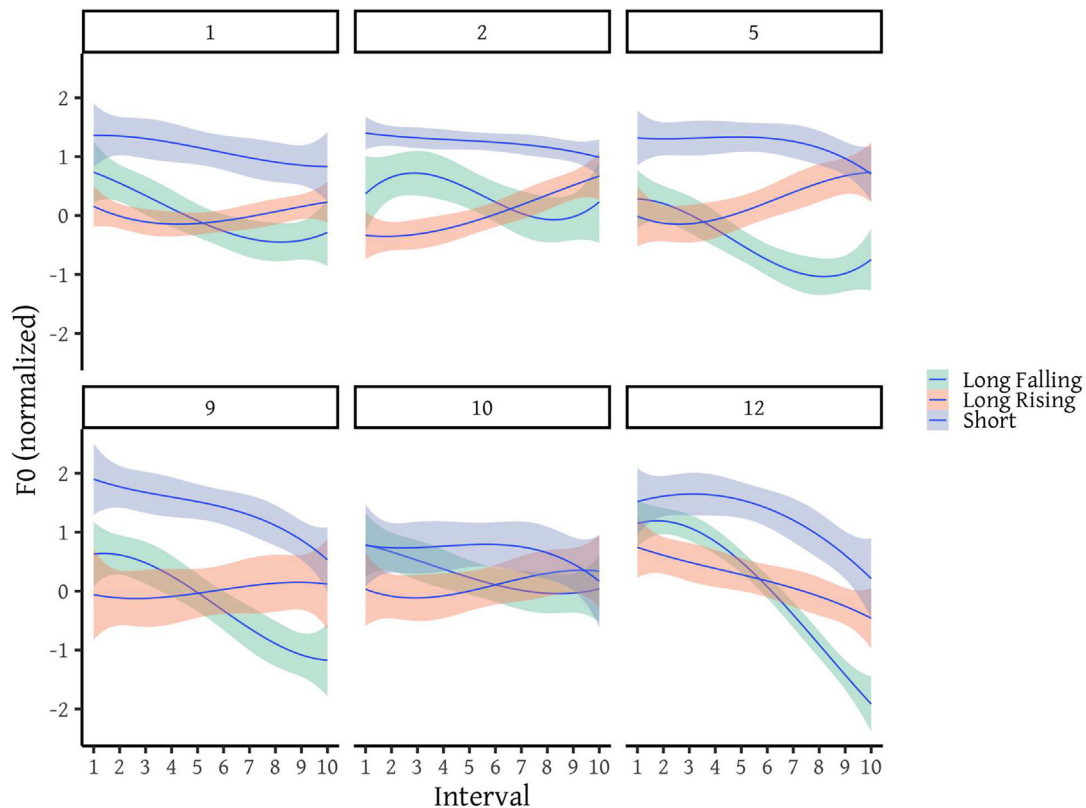


Fig. 3. Normalized pitch tracks for monosyllables in the elicited monosyllables for selected (numbered) participants.

ical effects. Additional acoustic parameters (duration, intensity) will be presented in Section 3.

2.4. Disyllables

In disyllables and longer words, stress also plays a role in the distribution of tone. As we will see, however, stress, vowel length, tone and vowel quality form complex interactions. In this section, we focus mainly on the phonological description of disyllabic prosodic patterns.

Most Žiri words are rather short (Stanonik, 1977). Žiri does not have epenthesis observed in some other varieties of Slovenian, but exhibits a number of deletions, historically speaking. As such, data from other varieties of Slovenian can act as a conservative estimate of word length in Žiri. Jurgec (2019) finds that 70% of stems in Šmartno Slovenian are monosyllabic. In a corpus study, Becker and Jurgec (2020) examine Standard Slovenian, which has 42% monosyllabic words, 51% disyllables and only 7% trisyllables or longer words. There is only a handful of native stems that are longer than

disyllabic (Jurgec, 2007), all others are loanwords. In this paper, we look at monosyllables and disyllables. The patterns described can be extended to longer words, but distributions in longer words are more limited (see also Stanonik, 1977).

Acoustically, stress is cued by higher pitch, increased duration, and higher intensity. Acoustic analysis of pitch is provided at the end of this section. A more detailed acoustic analysis, which includes duration and intensity, can be found in Section 3.

Initial long vowels. The disyllables can be divided into three groups by vowel length. While there can be only one long vowel per word, it can be initial or final, and there are also words without long vowels. Examples of disyllables with initial long vowels are provided in (4).

(4) Distribution of tone in disyllables with initial long vowels (Stanonik, 1977)			
Falling		Rising	
'skáakot	'to jump'	'meédvæt	'bear'
'stáarost	'age'	'eézer	'lake'
'záatost	'sadness'	'jeétfmen	'barley'
'yáajtrof	'rose'	'sreébor	'silver'

The long vowel in the initial syllable of these disyllabic words may have either a rising or a falling contour, just like monosyllables with a single long vowel. There is no tonal contrast in the second syllable, which we transcribe with a lack of High tone. Here all words end in a consonant, but the same pattern is observed in words ending in a vowel (e.g. ['bráado] 'chin, beard', ['zuéezdo] 'star' versus ['waádo] 'water', ['kaázo] 'goat'). The disyllables with initial long vowels conform to the generalizations established so far: long vowels must have a High tone (on one mora). These data further show that the posttonic short vowel can never bear a High tone. As we will see below, High tones are highly restricted in Žiri.

Final long vowels. A more complex situation is found in disyllables with the final long vowel (including diphthongs, which are always long). Four distinct patterns are attested (5).

(5) Distribution of tone in disyllables with final long vowels (Stanonik, 1977)			
a. Final stress			
Falling		Rising	
ze'líes	'iron'	zo'kleént	'to lock'
ko'léen	'knee'	ko'zouíts	'haystack'
po'léen	'log'	u'yaánt	'to guess'
ko'wáatf	'blacksmith'	o'traáts	'children'
b. Initial stress (unattested with [εε])			
Falling		Rising	
'kókóof	'chicken'	'kórwaáw	'bloody'
'uéjfer	'evening'	'pódoójk	'bolt washer'
'bótáan	'sick'	'lésién	'wooden'
'yótúop	'dove'	'ótruóp	'bran'

The first two groups of words have final stress on the long vowel (5-a), and this long vowel may have either a falling or rising tonal pattern, just like long vowels in monosyllables (2) and in the initial syllable of a disyllabic word (4). When stress falls

on a final long vowel, the initial short vowel is always toneless. The second two groups of words with final long vowels have stress on the initial short vowel (5-b). The initial short vowel bears a High tone, and a second High tone is realized on the second, long vowel. The two High tones can be adjacent, as in ['yótúop] 'dove', or non-adjacent, as in ['ótruóp] 'bran', in which the first mora of the long vowel lacks a High tone, but is flanked by High vowels on adjacent tone-bearing units.

The data in (5-b) are surprising because they contain unstressed long vowels, which also bear a High tone. Cross-linguistically, long vowels prefer stress and stress prefers long vowels (Prince, 1990; Hayes, 1995), and while both generalizations are true for the majority of Žiri words, this is not the case for all disyllables with final long vowels. From the perspective of the tonal patterns found in other dialects of Slovenian and the related BCMS, two tonal peaks observed in words like ['ótruóp] 'bran' are as far as we know unique. Historically, the Long Rising pattern is newer and arose via metaphony from falling tone when a posttonic vowel was deleted (Stanonik, 1977; Beguš, 2011). The existence of these disyllables allows us to revise our generalizations about Žiri: while stressed vowels and long vowels must have a High tone, long vowels can be unstressed if stress falls on the preceding short vowel.

The distinction between the groups with final (5-a) and initial stress (b) has to do with underlying tones. Recall that we model the difference between Long Falling and Long Rising monosyllables with two distinct autosegmental representations (3): a High tone can be linked to the first or second mora of a long vowel. The contrast between initial and final stress in disyllables with final long vowels (5) is crucial in determining the connection between tone and stress. This distinction suggests that words can have more than one underlying tone in Žiri. We represent the contrast between the two types of words with final long vowels by positing a different number of underlying High tones (6). In the words with final stress, there is only one underlying tone on the vowel. In the words with initial stress, however, there are two underlying tones, one realized on each syllable.

(6) Tone contrasts in disyllables with final long vowels	
Falling	Rising
$\begin{array}{c} \text{H} \\ \\ \text{kpleen} \\ \text{H} \quad \text{H} \\ \quad \\ \text{k} \quad \text{v} \quad \text{k} \quad \text{o} \quad \text{o} \quad \text{f} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{k} \quad \text{z} \quad \text{o} \quad \text{u} \quad \text{i} \quad \text{s} \quad \text{t} \\ \text{H} \quad \text{H} \\ \quad \\ \text{o} \quad \text{t} \quad \text{r} \quad \text{u} \quad \text{o} \quad \text{p} \end{array}$

Disyllables with long vowels lead us to conclude that stress in Žiri is predictable from tone: stress will fall on the leftmost of the two High tones.

Stanonik (1977) observes that prosodic patterns in Žiri are generally not affected by vowel quality: all long vowels (or diphthongs) can appear with either tone, in the stressed or immediately posttonic position. In (7), we list all six possible prosodic combinations based on the position of the long vowel (Initial or Final), pitch (Falling or Rising) and whether there is an additional initial High tone. The only exception is the long vowel [εε], which cannot appear in the posttonic position (Stanonik, 1977:301). This absence is peculiar, since the interactions

between vowel quality and prosody are very limited in the languages of the world. We return to the stress restriction on [ɛɛ] in Section 3.

(7) Possible and impossible disyllabic forms with long vowels ([ɛɛ] represents other vowels)					
		[ɛɛ]	[ɛɛ]	Real-word data	
Initial Long	Falling	'péetik	'péetik	(4)	
	Rising	'peetik	'peetik	(4)	
Final Long	Falling	pi'téek	pi'téek	(5-a)	
	Rising	pi'teék	pi'teék	(5-a)	
	Falling	Initial H	'píteek	'píteek	(5-b)
	Rising	Initial H	'píteek	'píteek	(5-b)

Words with Rising [ɛɛ] outnumber the words with Falling [ɛɛ] at least 10 to 1 in Stanonik (1977). The Falling [ɛɛ] typically appears in the final syllable of the word: monosyllabic [tʃɛɛl] 'forehead', [fɛɛjst] 'fine', [bɛɛj] 'flee-IMP.IMP', [tɛɛp] 'you-DAT'; disyllabic [stɔ'dɛɛnts] 'stream', [zɛ'lɛɛn] 'green-NEUT.SG.NOM'.³ This is grounded in historical developments in Žiri: [ɛɛ] arose due to retraction, and all such [ɛɛ] were rising; when the following vowel deleted because of another process, the High tone shifted one mora to the left, changing Long Rising [ɛɛ] into Long Falling [ɛɛ]. For a detailed historical explanation of these asymmetries, see Section 4.3.

No long vowels. The final group of disyllables has only short vowels (8). Only one realization is possible: stress is always initial, realized with a High tone.

(8) Distribution of stress and tone in disyllables with short vowels (Stanonik, 1977)		
'óku		'eye
'kúfer		'lizard
'stóbor		'pillar
'lípo		'linden tree

The attested pattern in disyllables with only short vowels is consistent with other data in this section in two respects: (i) all Žiri words thus must have a High tone on the stressed vowel, and (ii) if a short vowel has a High tone, it must be stressed, see (7). These two restrictions exclude most prosodic combinations in words with only short vowels (9). For instance, no disyllable with only short vowels can have two High tones (so *[pótó] is ungrammatical) or no High tones (*[poto]). The new generalization that disyllables without long vowels reveal is that the default position of stress is penultimate, ruling out final stress ([*pótó]). Longer words without long vowels also have penultimate stress ([tɔ'píne] 'shell, peel'). Below we argue that this preference for penultimate stress indicates trochaic footing.

(9) Possible and impossible disyllabic forms with only short vowels		
Possible	'poto	
Impossible	*pótó	(i. Short vowels with a High tone must be stressed)
	*poto	(ii. Stressed vowels must have a High tone)
	*potó	(i. and ii.)
	*pótó	(iii. Stress must be penultimate in words with only short vowels)
	*pótó	(i. and iii.)
	*pótó	(i., ii., and iii.)
	*pótó	(ii. and iii.)

Unlike the disyllables with long vowels, vowel quality in words with only short vowels is not restricted; in particular, the short [ɛ] can appear in the stressed (e.g. [mésu] 'meat', [vɛtʃer] 'evening', [lɛsién] 'wooden') or posttonic position ([plótʃet] 'pay', [mɛɛduet] 'bear', [kúfer] 'lizard').

Footing. In summary, stress in disyllables can be initial or final, which depends on the distribution of tone and vowel length. Stress is initial if High tone is present underlyingly on the first vowel. Otherwise, stress is final if that vowel is long or initial in words without long vowels. These restrictions on tone and stress are not accidental but make sense once footing is considered. In Žiri, feet are trochaic, which Žiri shares with both tonal and non-tonal Standard Slovenian (Jurgec, 2007; Jurgec, 2010; Jurgec, 2023). Moreover, trochaic feet are favored over iambic cross-linguistically and typologically (Hayes, 1995; Hyde, 2002; Kager, 2007). As shown in (10), disyllabic words with initial long vowels are entirely parsed into a disyllabic trochee. In words with initial short vowel, the footing depends on the distribution of tone: if the word only has a High tone on the long vowel, the foot is monosyllabic. If the word has two High tones, the foot is disyllabic. Finally, disyllables with only short vowels are footed in their entirety.

(10) Footing and prosodic contrasts in disyllables				
	Falling	Rising	Short	
Initial Long	H	H		Single High tone
	(ʔáalɔst)	(ʔeɛzer)		
Initial Short	H	H	H	
	kɔ('leɛn)	kɔ('zouts)	(ʔbku)	
	H H	H H		Two High tones
	(ʔkɔkuɔf)	(ʔɔtruɔp)		

This analysis accounts for four things. First, High tones are always footed. Second, the foot head must have a High tone. Third, long vowels must be footed. Fourth, the default position of the foot is at the end of the word, as seen in trisyllabic or longer words with only short vowels.

We do not make any assumptions about other feet in the word, which have no effect on our analysis (see Zec, 1999 for a parallel analysis of BCMS). Words with more than two syllables are rare, and while they pattern with disyllables, they do exhibit additional restrictions (see Section 3.2 for further discussion). In words with a long vowel, stress falls either on the long vowel (e.g. [kɔb'ɔasɔ] 'sausage', [ɣɔb'ɔakɔ] 'deep.FEM') or on the preceding short vowel (e.g. [pɔ'mɔraánʔɔ] 'orange', [tɛlɛ'viziʔɔ] 'television'). When the long vowel is ante- or preantepenultimate,

³ There are a few examples of non-final Long Falling [ɛɛ], all in the position before [j], such as the disyllabic [mɛɛjɛt] 'mix-INF.IMP' and trisyllabic [ɔ'bɛɛjɛt] 'hang-INF.IMP'. All of these examples involve [ɛɛ] in the position before tautosyllabic [j], where the contrast between the two front mid vowels is neutralized regardless of pitch. The quality of [ɛɛ] before tautosyllabic [j] is slightly different (higher) than elsewhere, which is also found in other varieties of Slovenian (for Standard Slovenian, see Jurgec, 2007 and Toporišič, 1976/2000).

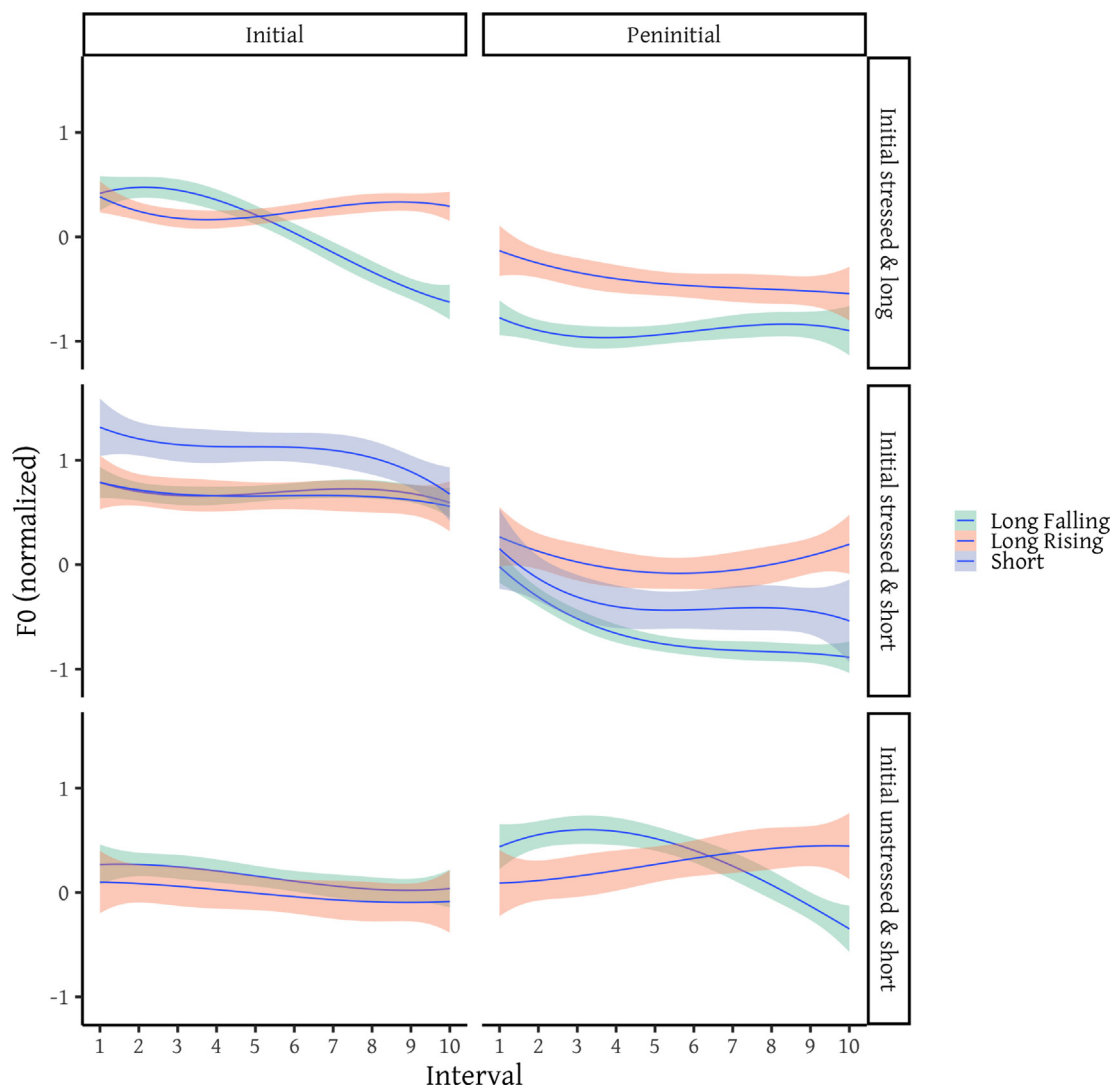


Fig. 4. Normalized pitch tracks in the elicited disyllables by the position of the vowel (initial and peninitial/final), prosodic combination (stress and quantity of the initial vowel), and pitch (Long Falling, Long Rising, and Short; marked in colors). At the top panels, the Long Falling forms include [króawo] ‘cow’ and Long Rising [eézer] ‘lake’ (4). In the middle, the Long Falling forms include [kókóoj] ‘chicken’, Long Rising [kórwaaw] ‘bloody’ (5-b) and Short [lípó] ‘linden tree’ (8). At the bottom, the Long Falling forms include [ko’léen] ‘knee’ and Long Rising [o’traáts] ‘children’ (5-a).

there are only two possibilities and both result in Long Rising contour on the long vowel: word stress can either be on that vowel ([‘táástutso] ‘swallow’, [dɔr’wáárntso] ‘woodshed’) or preceding short vowel ([kʰtúʃtááwntso] ‘lock’). Antepenultimate stress will need to be studied in subsequent work (for further discussion, see Stanonik, 1977:297; Beguš, 2011:21). In words with only short vowels, stress is penultimate ([tɔpíne] ‘husk’, [bɔrdo’vítso] ‘wart’), suggesting that feet are generally aligned with the right edge of the word, but this can be overruled by the requirement for long vowels to be footed.⁴ As with disyllables, no trisyllabic or longer words have posttonic [ɛɛ]. We

build upon these generalizations in Section 3, which explores prosodic alternations.

Phonetic realization of tone. The final issue to resolve in this subsection is the phonetic realization. As we have seen in this section, the disyllables contrast seven different prosodic combinations depending on stress (initial or final), quantity (initial long, final long or only short) and for long vowels also on whether the pitch is falling or rising. We present all seven combinations in Fig. 4. The panels on the left present the pitch of the first (initial) vowel whereas the panels on the right present pitch tracks of the second (peninitial) vowel. We choose the term ‘peninitial’ over ‘final’ to avoid ambiguity with monosyllables where initial vowels are also final, as we will further discuss in Section 3. The top two panels present words with initial stressed and long vowels, while the bottom two panels present words with final stressed and long vowels. The middle two panels present words with initial stressed and short vowels. The colors allow us to distinguish the three different pitch accents. ‘Long Falling’ and ‘Long Rising’ refer to the realization on the long vowel—regardless of whether it is stressed. ‘Short’ refers

⁴ Among polysyllables with only short vowels, Stanonik (1977) lists 11 words that have final stress, all of which represent exceptional paradigms or forms (e.g. [mɛˈʃi] ‘mouse.GEN.PL’, [s kɔstˈmɪ] ‘with bone.INSTR.PL’, [pɔˈkúst] ‘try.INF’, [vɔˈnɪl] ‘drive.PART.PL.M’, [lɛˈtɪ] ‘fly.3SG’). These forms are also phonologically exceptional in the standard (Jurgec, 2007; Toporišič, 1976/2000) and other dialects of Slovenian (Božič, 2015; Jurgec, 2019). Furthermore, there is one variant form with antepenultimate stress [mɪsnɛjɛ ~ ‘mɪsnjɛ ~ ‘mɪsnɛjɛ] ‘think.3PL’. Over 90% of polysyllables with only short vowels in Stanonik (1977) have penultimate stress. Prosodic alternations within paradigms will need to be studied more fully in subsequent work.

to words with only short vowels, which always have initial stress in disyllables; they are presented only on the middle two panels.

Mirroring the data for monosyllables, we see a clear tonal contrast for all three pitch accents for each combination of stress and quantity, though not necessarily for each vowel.

The top two panels present disyllables that have an initial long vowel, which is always stressed, followed by a short vowel. The colors are consistent for both vowels even though the phonological representation is not necessarily distinct on each of the two vowels. For instance, the pitch of the Long Falling [*kráaw*] ‘cow’ is presented on both panels: the initial vowel [*áa*] is on the left, whereas the peninitial vowel [*a*] is on the right. Conversely, the pitch of the Long Rising [*éézer*] ‘lake’ is split into the initial vowel [*éé*], included on the left, and peninitial vowel [*e*] included on the right.

The initial, long vowel clearly distinguishes two different contours: Long Falling pitch has a tonal peak at the beginning, whereas the Long Rising pitch has it towards the end. We capture this distinction by positing a High tone on the first mora of the initial vowel for Long Falling pitch (i.e., [*kráaw*] ‘cow’) as opposed to the second mora for the Long Rising pitch ([*éézer*] ‘lake’). The two pitch accents converge at the peninitial, short vowel, although the Long Rising pitch starts much higher than Long Falling. This is a phonetic rather than a phonological effect: both pitch accents are represented identically as lacking a High tone on the peninitial vowel.

The middle two panels display words with initial stress, whereas the long vowel is peninitial (e.g. [*kókóo*]). Words with only short vowels ([*líp*] ‘linden tree’) are also included in the two panels. For all three pitch accents, the highest pitch is on the initial vowel, which is consistent with having a High tone on that vowel. The peninitial vowel can be distinguished between Long Falling and Long Rising pitch accents, with the latter ending much higher than the former. We interpret this as Long Falling words having a second High tone on the first mora of the peninitial vowel (e.g. [*kókóo*] ‘chicken’), whereas Long Rising words have the High tone on the second mora ([*kórwaáw*] ‘bloody’). Words with only short vowels have slightly higher pitch on the initial, stressed vowel, which is consistent with the data for monosyllables. On the peninitial vowel, the Short pitch is between both long pitches, which we interpret as a lack of a High tone ([*líp*] ‘linden tree’). This is an artefact of how we represent the data here: the pitch intervals are identical regardless of duration. In reality, short vowels are shorter than long vowels (see Fig. 7 for more detail).⁵

The bottom two panels show words with long vowels on the peninitial vowel, which is also stressed (e.g. [*ko’léen*] ‘knee’). First, focusing on the initial vowel (i.e., the bottom left panel),

we see that the pitch is comparatively much lower when compared to the words with initial stress (i.e., the panel above). We interpret this distinction as the absence (bottom left panel) versus the presence (middle left panel) of a High tone. Moving on to the peninitial vowel (bottom right panel), we see that the Long Falling pitch peaks at the third interval, which we interpret as a High tone on the first mora of that vowel (e.g. [*ko’léen*] ‘knee’). The Long Rising pitch, in contrast, raises at the end, which we interpret as a High tone on the second mora ([*p’traáts*] ‘children’). Note that this paper focuses on words with initial stress, which means we had fewer tokens of disyllables with final stress, particularly Long Rising (see Table 3 in the Appendix for complete token counts).⁶

We can also compare the different disyllabic prosodic combinations in disyllables in another way. Fig. 5 groups panels by pitch accent rather than by position. The top two panels present Long Falling pitch, which can have either initial or peninitial stress, and with a long vowel on the first or second vowel; these prosodic combinations are presented by colors. Words with initial long and stressed vowels are marked in magenta. The peak is on the first mora of the initial vowel, which we interpret as a High tone on that vowel followed by an unstressed peninitial vowel that lacks a High tone (e.g. [*kráaw*] ‘cow’). In blue are words with initial stress but peninitial long vowels. High pitch on the initial vowel corresponds to a High tone on that short vowel, whereas the fall on the peninitial vowel corresponds to the High tone on the first mora of the peninitial vowel (as in [*kókóo*] ‘chicken’). This contrasts with words in green that have initial short vowels and peninitial long vowels, but the stress is on the peninitial vowel (e.g. [*ko’léen*] ‘knee’). The lower pitch on the initial syllable is interpreted as a lack of a High tone to distinguish it from the words with initial stress, which have either High tone on the first mora (as in the case of initial long vowels, in magenta) or only mora (as in the words with initial short vowels, in blue). Moving to the top right panel and continuing to examine the data in green, we see that the pitch peaks towards the beginning, which we interpret as a High tone on the first mora of the peninitial vowel.

The Long Rising pitch is next, presented in the middle two panels. We first examine the words with initial stressed, long vowel followed by an unstressed, short peninitial vowel (magenta). The pitch peaks towards the end of the initial vowel, which corresponds to a single High tone on the second mora of the initial vowel (e.g. [*éézer*] ‘lake’). The pitch of the words with initial stress but a final long vowel (blue) has a higher pitch on the first vowel, which we interpret as having a High tone on that vowel (e.g. [*kórwaáw*] ‘bloody’). On the peninitial vowel, the pitch increases towards the end, which we interpret as indicative of a second High tone on the last mora. The words with peninitial long vowels can also have peninitial rather than initial stress (green). When comparing the two groups of words, we see that those with peninitial stress (in green) have

⁵ An anonymous reviewer questions the analysis of the Long Falling pitch as having a phonological High tone on the first mora of the peninitial vowel. Under this alternative, [*kókóo*], not [*kókóo*] ‘chicken’ should be a more accurate transcription. The acoustic data shows that the Long Falling pitch has the lowest normalized F0 in the second half of the peninitial vowel. To capture the difference between the first and second mora of the peninitial vowel, we propose that the first mora has a High tone while the second mora lacks it. This is exactly the opposite of what we see in the Long Rising pitch. This also accurately captures the difference between Long Falling ([*kókóo*] ‘chicken’) and Short pitch ([*líp*] ‘linden tree’): the first vowel has a High tone in both, yet pitch starts to fall earlier in Short pitch words, while remaining flat until the end of the vowel in Long Falling pitch words. This longer plateau indicates a High tone on the first mora of the Peninitial vowel. We also note that we could not tightly control for the effect of consonants, syllable structure, vowel quality, and speaker, which can lead to some variability across the pitch accents.

⁶ The Long Rising pattern is much rarer than the Long Falling due to how it arose historically. Upon closer examination of the Long Rising elicitation we segmented, it turned out that 30 out of 38 tokens were not consistently pronounced with the expected Long Rising contour (Stanonik, 1977) but appeared variable (Long Rising or Falling). Due to an abundance of caution, we excluded these ambiguous tokens. The pink dataset in the bottom two panels of Fig. 4 is thus based on 8 tokens, yet the contour is reliably distinct from other prosodic combinations. Including all 38 tokens produces a pitch track that is flatter but nevertheless raises slightly towards the end.

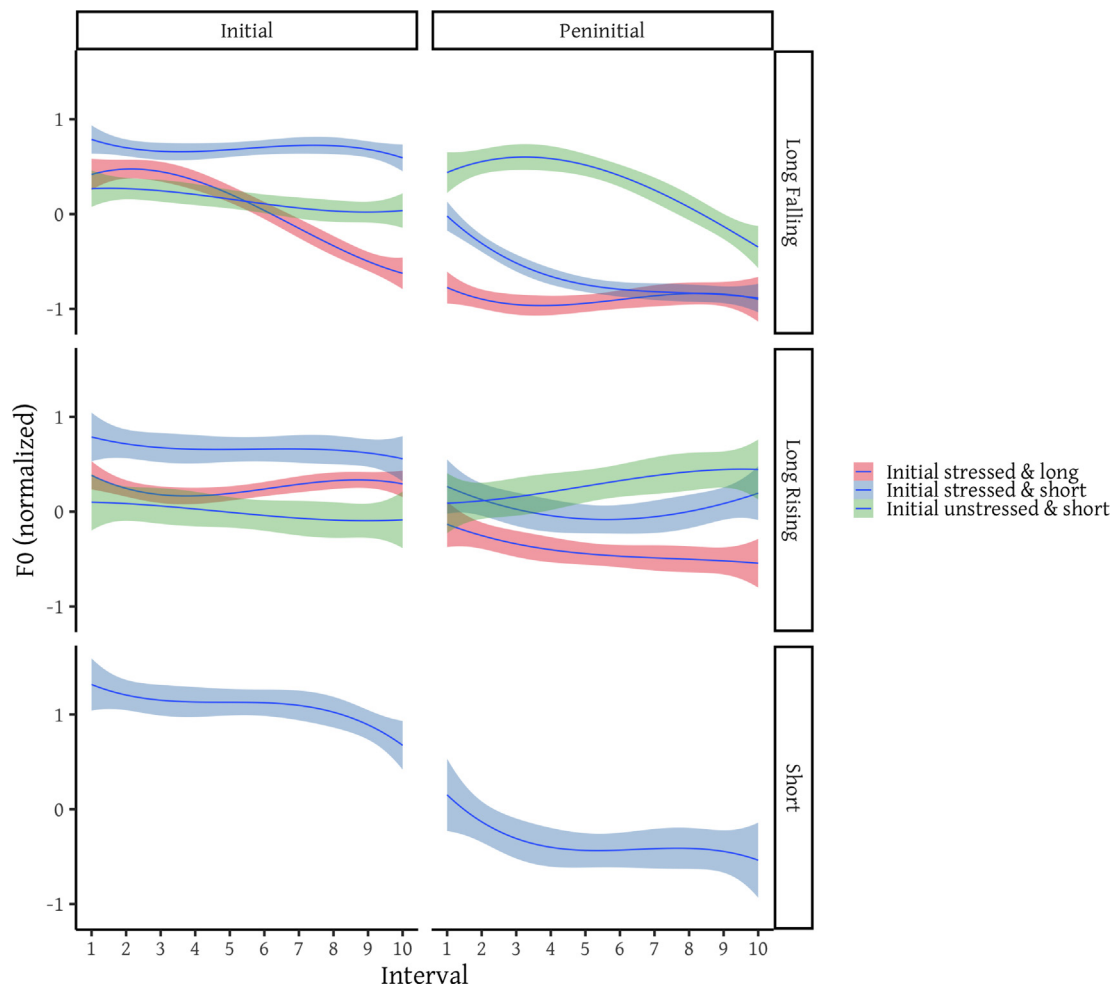


Fig. 5. Normalized pitch tracks in the elicited disyllables by the position of the vowel (initial and peninitial/final), pitch (Long Falling, Long Rising, and Short) and prosodic combination (stress and quantity of the initial vowel; marked in colors). At the top panels, the Long Falling forms include those with initial stress and long vowels [kráowa] ‘cow’ (4), initial stress and peninitial long vowels [kókóo] ‘chicken’ (5-b) and final stress [kóléen] ‘knee’ (5-a). In the middle, the Long Rising forms include those with initial stress and long vowels [éézer] ‘lake’ (4), initial stress and peninitial long vowels [kórwaáw] ‘bloody’ (5-b) and final stress [o’troáts] ‘children’ (5-a). At the bottom, the Short forms include disyllables like [lípó] ‘linden tree’ (8). Stress is always initial in Short disyllables.

a much lower pitch on the initial vowel compared to the words with initial stress (in blue). This suggests that words with peninitial stress do not have a High tone on the initial vowel (e.g. [o’troáts] ‘children’), which is characteristic of words with initial stress. The peninitial vowel for both contours is a natural continuation of the initial vowel. This is a phonetic effect. Both contours, however, end with a rise, suggesting a High tone on the final mora.

Finally, the bottom two panels present words with only short vowels. These always have a higher pitch on the initial vowel, indicating a High tone (and stress) on that vowel (e.g. [lípó] ‘linden tree’).

2.5. Interim summary

In this section, we presented the distribution of vowel length, tone, stress, and vowel quality in Žiri. These properties are often restricted and interact with one another in complex ways. We summarize the key descriptive generalizations in (11).

- (11) Tone, stress, vowel length, and vowel quality in Žiri
 - a. High tone is lexically specified and contrastive; there is no phonological Low tone.
 - b. Every foot, and thus every word, has at least one High tone.
 - c. Stress falls on the leftmost High tone within the word. The leftmost High tone coincides with the left edge of a trochee.
 - d. There is at most one long vowel per word.
 - e. Long vowels (and diphthongs) must be footed, and will always have a High tone, either on the first (Falling) or second mora (Rising).
 - f. Short vowels with a High tone are always stressed, and short vowels without a High tone are always unstressed.
 - g. In words without long vowels and underlying High tones, the trochaic foot is rightmost, assigning penultimate stress and High tone.
 - h. Long [ɛɛ] is always stressed, while other long vowels can be stressed or unstressed (posttonic).

These distributions are also observed in alternations, which we turn to next.

3. Retraction

So far we have seen the realization of tone in isolated words and short phrases or sentences. In this section, we look at the alternation affecting words that are preceded by a clitic. In Slavic languages, stress is often retracted from the content word to the preceding clitic (see [Franks & King, 2000](#) for a comprehensive overview). Žiri shows complex pitch accent contrasts, so retraction is more complex than in other Slavic languages.

Cross-linguistically, clitics can have a variety of shapes (see [Anderson, 2005](#); [Anderson, 2011](#) for a cross-linguistic overview and [Zwicky & Pullum, 1983](#) for English). In Žiri, however, clitics are monosyllabic, stressless, toneless, and contain only short vowels (as in Standard Slovenian, see [Toporišič, 1976/2000](#); [Jurgec, 2007](#)). Put differently, proclitics form a Prosodic Word with the following content word ([Nespor & Vogel, 1986](#); [Peperkamp, 1997](#); [Dixon & Aikhenvald, 2002](#)). In terms of word class, the majority of clitics reported in this paper are prepositions. The clitics that appear in our data are listed in (12). The complete inventory of clitics in Žiri is a larger, but closed class. All clitics used in the experiment (Section 4) are prepositions.

(12)	Žiri clitics		
	Prepositions	no	'on, onto, in, into'
		po	'to, after, on'
		do	'to, until'
		zo	'for, around, behind'
		u	'in, into'
		pod	'under, below'
	Adverb	že	'already'
	Copula	je	'is'
	Conjunction	pa	'and, but'

3.1. Monosyllables

In Žiri, words with initial stress show an alternation when preceded by a clitic. As we will see, the mechanism behind this alternation is dependent on the length of the word. In particular, the clitic functions as an additional syllable. This means that the patterns observed in monosyllables with clitics are nearly identical to disyllables without clitics, and the patterns observed in disyllables with clitics are identical to trisyllables without clitics.

The data in (13) compare monosyllabic nouns with and without preceding clitics. Without clitics, the monosyllables distinguish three different prosodic patterns, as already seen in (2). The proclitics have a High tone. In Žiri stress falls on the leftmost High tone (Section 2.4), and since proclitics are part

of the same prosodic word as the content word, the stress retracts to the clitic. At the same time, monosyllables with Long vowels retain their tonal contours, while those with short vowels lose their High tone because it is not possible on unstressed short vowels.

(13) Retraction in monosyllables with proclitics				
a. Long Falling	núos	'nose'	'nó núos	'on (the) nose'
	wráat	'neck'	'zó wráat	'around (the) neck'
	méex	'bellows'	'zó méex	'for (the) bellows'
b. Long Rising	yuópts	'muzzle'	'nó yuópts	'on (the) muzzle'
	oótk	'little eye'	'ú oótk	'into (the) little eye'
	jaópk	'apple'	'pó jaópk	'after (the) apple'
	xleépts	'loaf'	'pó xleépts	'after (the) loaf'
c. Short High	króx	'bread'	'pó krox	'to (the) bread'
	kóp	'pile'	'nó kop	'on/onto (the) pile'
	měj	'mouse'	'pó mej	'after (the) mouse'

The prosody of the monosyllables with proclitics in (13) resembles that of disyllables with initial stress. As we have seen in (5), disyllables with long vowels in the second syllable distinguish a falling and a rising contour. When the first syllable has a High tone, stress is initial, just as we have observed with monosyllables with proclitics. For instance, a disyllabic word such as ['kókóof] 'chicken' has similar distributional properties as a monosyllabic word preceded by a clitic, such as ['nó núos] 'on (the) nose'; in this particular case, both have stress on the initial short, High tone vowel, followed by a long vowel with a High on the first mora (Falling). In line with the practice in the Slavic and Slovenian literature ([Stang, 1957](#); [Sussex & Cubberley, 2006](#); [Beguš, 2011](#); [Kapović, 2015](#)), we refer to this alternation as 'retraction', even though retraction in Žiri is mediated by High tone.

There are two additional pieces of evidence for direct parallels between the two disyllabic forms (monosyllables with proclitics and disyllables without). First, consider monosyllables with short vowels in (13-c), which lack a High tone on the content word. This is identical to the prosody of disyllables without long vowels, as in (8), all of which have a High tone and stress on the initial syllable, followed by a syllable without High tone. Thus, the disyllabic ['óku] 'eye' has the same prosody as the retracted monosyllabic ['pó krox] 'to (the) bread'. Alternations of short monosyllables are reported and tested for the first time in this paper.

The other piece of evidence comes from a second set of monosyllables. Recall that among disyllables with a final long vowel some have initial stress (and an additional High tone) while others have final stress. Monosyllables preceded by clitics show similar behavior: while some show retraction (13), others do not (14), and are unaffected by proclitics. For instance, while the Long Falling monosyllable [núos] 'nose' is preceded by a clitic that has a High tone ['nó núos] 'on (the) nose' showing stress retraction, the Long Falling monosyllable [púot] 'path' retains stress on the stem, with the clitic surfacing without a High tone [no 'púot] 'on (the) path'.

(14) Lack of retraction in monosyllables with proclitics

a. Long Falling	yréex	'sin'	ʒe 'yréex	'already sin'
	miést	'town'	u 'miést	'in town'
	púot	'path'	no 'púot	'on (the) path'
b. Long Rising	tʃéel	'forehead'	po 'tʃéel	'on (the) forehead'
	kaásts	'reaper'	po 'kaásts	'but (the) reaper'
	jeésk	'tongue'	zo 'jeésk	'for (the) tongue'
c. Short High	xó rpt	'back'	zo 'xó rpt	'behind (one's) back'
	wós	'village'	no 'wós	'in (the) village'
	tútʃ	'light'	po 'tútʃ	'into (the) light'
	sén	'hay'	po 'sén	'after (the) hay'

How can we make sense of this difference between the monosyllables with retraction and those without? A key generalization is that all clitics behave uniformly in that their prosody is fully predictable from the following word. For instance, the monosyllable [núos] 'nose' can be preceded by any clitic and it shows the same pattern regardless: [pó núos] 'after, on top of', [zó núos] 'for', [ʒé núos] 'already', [jé núos] 'is' etc. Conversely, all clitics preceding [púot] 'path' will remain stressless: [po 'púot] 'on, following', [zo 'púot] 'for, after', [ʒe 'púot] 'already', [je 'púot] 'is'. In fact, the same is true for all content words and all clitics. This suggests not only that the prosody is determined by the monosyllabic content word rather than the clitic, but also that the distinction between retracting and non-retracting monosyllables is not predictable and has to be specified lexically.

Recall that stress in Žiri follows from the distribution of High tone. In this sense, calling the alternation '(stress) retraction' (Stanonik, 1977; Beguš, 2011) is misleading, as it is actually the High tone distribution that is different, as observed in disyllables without clitics. We present a unified analysis of both types of disyllabic forms (i.e. disyllables without clitics and monosyllables with clitics), as the patterns are largely the same.

The distinction between the retracted and non-retracted stems comes from their tonal representation. As we have seen in Section 2.4, the disyllables differ whether they have stress on the long vowel or the preceding short vowel. The latter group had two underlying High tones. We suggest that the same mechanism is at play with clitics. The stems that show retraction have an additional High tone, which is not associated with a particular position in the monosyllable, but is instead floating (Odden, 1988; Akinlabi, 1996). The floating tone docks onto the clitic when possible, but is not realized otherwise. This allows us to distinguish the retracting and non-retracting monosyllables. The mappings are shown in (15). Unlike the distributions in Section 2, the existence of alternations allows us to distinguish from input/underlying and output/surface forms. Note that the very same mapping would also work for disyllables with final long vowels.

(15) Monosyllables with long vowels can have a floating High tone

	Input	Output
Retraction	$\begin{array}{c} \text{H H} \\ \quad \\ \text{nd sook} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{('nd sook)} \end{array}$
No retraction	$\begin{array}{c} \text{H} \\ \\ \text{nd puot} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{nd ('puot)} \end{array}$

Returning to the data in (14), notice that short vowels display two distinct patterns, much like long vowels: the first group shows an alternation, while the second retains stress on the content word. This differs from the situation in short disyllables without clitics, where only one pattern is attested: disyllabic and longer words without long vowels have predictable stress on the penultimate vowel (8). Moreover, unlike the retracted monosyllables with long vowels, short vowels do not retain the High tone on the monosyllable. These patterns suggest that the analysis of short vowels is more complex. One possible analysis is parallel to (15): the retracting short vowel nouns have a floating High tone in addition to the linked one. The floating tone docks to the clitic, as with long vowels. Additionally, the High tone on the footed, posttonic short vowel delinks. This is because Žiri does not allow High tones on unstressed short vowels. An alternative analysis is also possible, with short vowel roots lacking an underlying linked tone, which is then assigned a High tone because all stressed syllables must have a High tone. The retracted monosyllables still have a floating tone which docks on the clitic. Regardless of which analysis we adopt, floating tones are used to capture retraction, just as with monosyllables with long vowels.

When it comes to vowel quality, the only restriction is regarding the long [ɛɛ]: while Long Rising [ɛɛ] is considerably more common than Long Falling [ɛɛ], neither allow stress on the clitic. This means all words with [ɛɛ] fall in the non-retracted group (14). This does not extend to short vowels: while [pó mɛʃ] 'after (the) mouse' displays retracted stress, [po 'sén] 'after (the) hay' does not.

To check for these generalizations, we considered the entirety of Stanonik's (1977) data, which consists of 193 monosyllables. The vowels are not equally represented. For instance, [aa] is the most frequent long vowel, while [oo] is the least frequent one; CV(V) C monosyllables are more common than CV(V) and CV(V) CC monosyllables. Nevertheless, it seems that retraction is possible from all vowels and regardless of syllable structure, with a single exception, the long vowel [ɛɛ] (as noted explicitly by Stanonik, 1977:301). None of the words with [ɛɛ], regardless of tone, show retraction. This absence is peculiar, since the interactions between vowel quality and prosody are very limited in the languages of the world. While there is phonetic grounding for a connection between vowel quality and tone via intrinsic fundamental frequency or between stress and sonority, this is not a possible interpretation of the Žiri data: it is only [ɛɛ] that lacks retraction, but not the remaining vowels. Some of these vowels are higher [ii, uu, ee, oo, ie, uo], whereas others are lower [aa] (the reverse is true for sonority). As such, it does not appear that the prosodic restriction on [ɛɛ] is phonetically motivated. Moreover, the short [e] has no restriction and can appear retracted or not. The key question of this paper is whether this absence is solely an artifact of the data sample. In Section 4, we ask whether this gap is systematic in the grammar of Žiri speakers or solely lexical. Experimental data confirm that the gap is in fact systematic.

To support these transcriptions of retracted and non-retracted monosyllables, we present normalized pitch tracks. In Fig. 6, the panels on the left present pitch tracks on the clitic vowel, whereas the right presents the tracks on the stem monosyllable. The top two panels present non-retracted realizations, whereas the bottom two panels present retracted

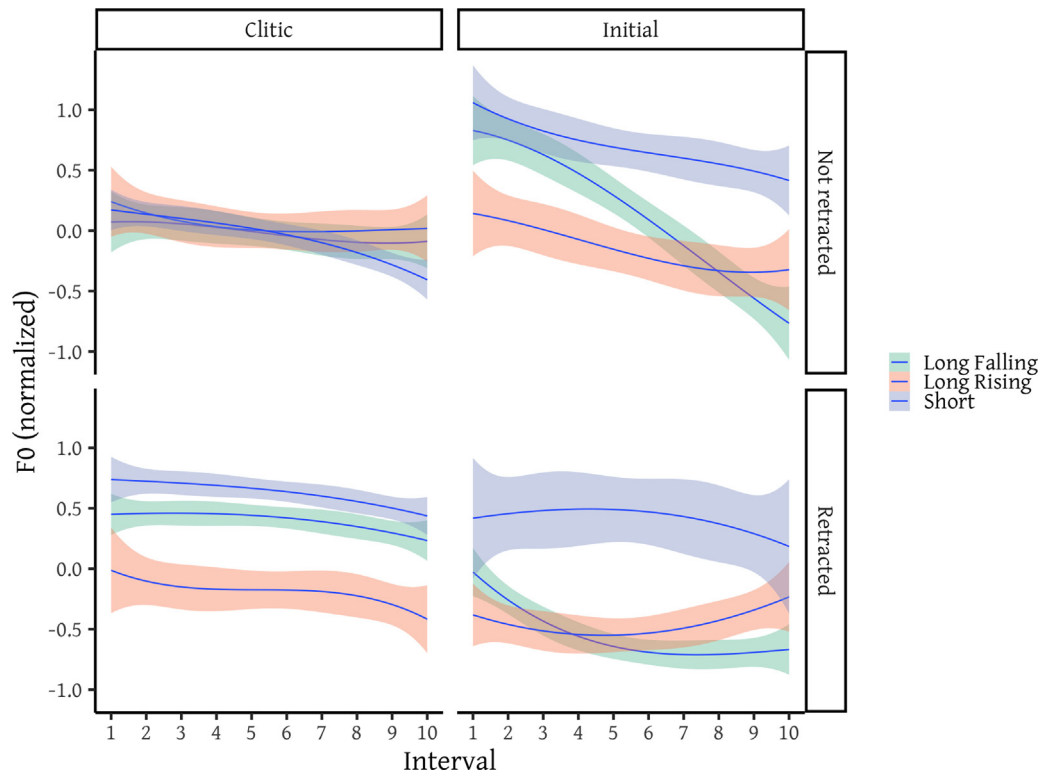


Fig. 6. Normalized pitch tracks in the elicited monosyllables by the position of the vowel (clitic, initial or only vowel of the monosyllabic stem), retraction, and pitch (Long Falling, Long Rising, Short). At the top, the non-retracted forms include Long Falling [ʒe 'vréex] 'already sin' (14-a), Long Rising [zo 'jeésk] 'for (the) tongue' (14-b), and Short [po 'sén] 'after (the) hay' (14-c). At the bottom, the retracted forms include Long Falling [nó núos] 'on (the) nose' (13-a), Long Rising [ú oóŋk] 'into (the) little eye' (13-b), and Short [pó krox] 'to (the) bread' (13-c).

realizations. The non-retracted realizations have lower pitch on the clitic when compared to the stem vowel, corresponding with our interpretation of the lack of High tone on the clitic as well as the presence of High tone on the stem vowel. The Long Falling pitch has a peak at the beginning of the stem vowel. The Long Rising pitch has a much lower realization and is phonetically falling. The slight rise at the end is not as sharp as we would expect given the phonological interpretation, yet the overall pitch is crucially still distinct from the Long Falling. We interpret the distinction between Falling and Rising contours as having a High tone on the first or the second mora, respectively. Short vowels have high pitch throughout the stem vowel even though there is an overall falling trend.

The retracted realizations look quite different. All three pitch accents have a higher realization on the clitic when compared to the stem vowel, which we interpret as having a High tone on the clitic. Moreover, the Long Falling pitch falls throughout the stem vowel, whereas Long Rising is U-shaped, with an increase towards the end of the stem vowel (as in the non-retracted realizations). Although these retracted stem-vowel realizations are slightly different phonetically than the non-retracted realizations, we interpret them as having the same phonological representations, with the High falling on the first mora of the Long Falling pitch and the second mora of the Long Rising pitch.

The realization of retracted short vowels is also distinct from non-retracted, with the clitic having the highest pitch in the retracted condition. We represent the non-retracted realization as lacking a High tone on the clitic, while the retracted version has a High tone on the clitic. As for the stem vowel, the higher pitch in the non-retracted condition is consistent with a High

tone. We interpret the non-retracted realization as lacking a High tone on the stem vowel, with the higher pitch on the stem vowel being a phonetic effect, but the evidence for this is inconclusive: the large confidence intervals suggest substantial variation across participants. Short vowels need to be further investigated in subsequent research.

The pitch tracks of monosyllables in proclitic environments in Fig. 6 are comparable to disyllables in the non-clitic environments, seen in Fig. 4. The non-retracted forms in Fig. 6 are comparable with disyllables with final stress in Fig. 4, while the retracted forms are comparable with disyllables with initial stressed and short vowels. The disyllables with initial long vowels do not have an equivalent in this section, as all clitics have short vowels.

3.2. Disyllables

Stress can also retract from disyllables and longer words with an initial long vowel, which constitute the core of the data in this paper. As we will see, the disyllables display several restrictions, the productivity of which will be tested experimentally in Section 4.

The disyllabic data is shown in (16). As with monosyllables, we see two general patterns: some words show retraction (a), while others do not. There are, however, a few differences. First, short vowels never show retraction in disyllables. A disyllable with two short vowels and two High tones will be realized with a trochee aligned with the right edge, with the penultimate syllable bearing stress and High tone, while the clitic and final syllable will be unstressed and lacking a High tone. In other

words, the foot in words without long vowels is always aligned with the right edge of the word, regardless of the number of syllables or the presence of a clitic. Thus, monosyllables can show retraction or not, while still maintaining the alignment of the foot with the right edge in either case; in contrast, disyllables will always have penultimate stress as retracting stress to the antepenultimate syllable would require the foot to no longer be aligned with the right edge of the word. Second, the Long Falling disyllables become rising after retraction. For instance, 'head-ACC' is falling [ʔááwɔ], but attains a rising contour with retraction [nó ʔááwɔ] 'on (top of the) head-ACC', which crucially differs from monosyllables which retain the falling contour [nó núos] 'on (the) nose', as in (13). These Long Falling disyllables present the only instance where we see a tonal alternation on the long vowel.

(16) Disyllables with proclitics

a. Retraction

Long Falling	ʔááwɔ	'head-ACC'	nó ʔááwɔ	'on'
	brááɔɔ	'beard'	zó brááɔɔ	'for'
Long Rising	xruúʔkɔ	'pear'	pó xruúʔkɔ	'after'
	eéɛɛ	'lake'	nó eéɛɛ	'on'

b. No retraction

Long Falling	brééɛɛ	'birch tree'	pɔɔ brééɛɛ	'under'
	krááwɔ	'cow'	pɔ krááwɔ	'after'
Long Rising	kaáɔɔ	'scythe'	pɔ kaáɔɔ	'after'
	kaáɔɔ	'goat'	nɔ kaáɔɔ	'on'
Short High	xíʔe	'house'	u xíʔe	'in'
	lípɔ	'linden tree'	nɔ lípɔ	'on'

The generalization we can draw from Long Falling disyllables is that two adjacent High tones are not allowed in non-final syllables. This is also consistent with the prosody of trisyllabic and longer words, which cannot have a falling tone unless the long vowel is word-final. While there are Long Rising words

like [tɛɛ'viziipɔ] 'television', the mirror Long Falling words are entirely unattested, *[tɛɛ'viziipɔ]. Another convincing piece of evidence comes from suffixed disyllabic stems. Final Long Falling stems with initial stress, such as [vɛʔʔiɛr] 'evening', become Long Rising when followed by a suffix, so not *[vɛʔʔiɛrɔ] but [vɛʔʔiɛrɔ] 'GEN'; Long Rising disyllables do not show this alternation. We elaborate on this restriction on the distribution of multiple High tones in Section 4.3. In contrast, the Long Rising pitch is maintained in retracted forms.

To summarize, disyllables with clitics differ from monosyllables in two key aspects: the Long Falling words neutralize with Long Rising when stress is retracted, and words with only short vowels never show retraction. In all other respects, the generalizations are identical: retraction is realized as High tone and stress on the clitic and the distinction between retracted and non-retracted words is lexical. Stanonik (1977) does not elaborate on the distribution of retraction. In particular, monosyllables appear equally likely to show retraction as disyllables, and falling pitch retracts at rates similar to rising pitch. When it comes to vowel quality, none of the known cases of disyllables with [ɛɛ] show retraction (e.g. [zɛɛmɛ] 'soil, earth' ~ [u zɛɛmɛ] 'in (the) soil, earth'; [sɛɛdɔ] 'saddle' ~ [nɔ sɛɛdɔ] 'in (the) saddle'), which is similar to monosyllables. All other vowels display at least some items with retraction. The vowel quality restriction, which lacks clear phonetic grounding, will be explored in Section 4.

Retraction in disyllables will form the core of our experiment, so we now turn to the acoustic analysis to support our transcriptions. In particular, we compare forms with and without retraction, while examining duration, intensity, and pitch tracks. The forms analyzed here all have initial stress when in isolation. Fig. 7 shows duration for each vowel (clitic, stem initial and peninitial) depending on vowel length and retraction. The left-hand panel presents the words in the non-retracted condition, such as [pɔ krááwɔ] 'after (the) cow' (16-b) and

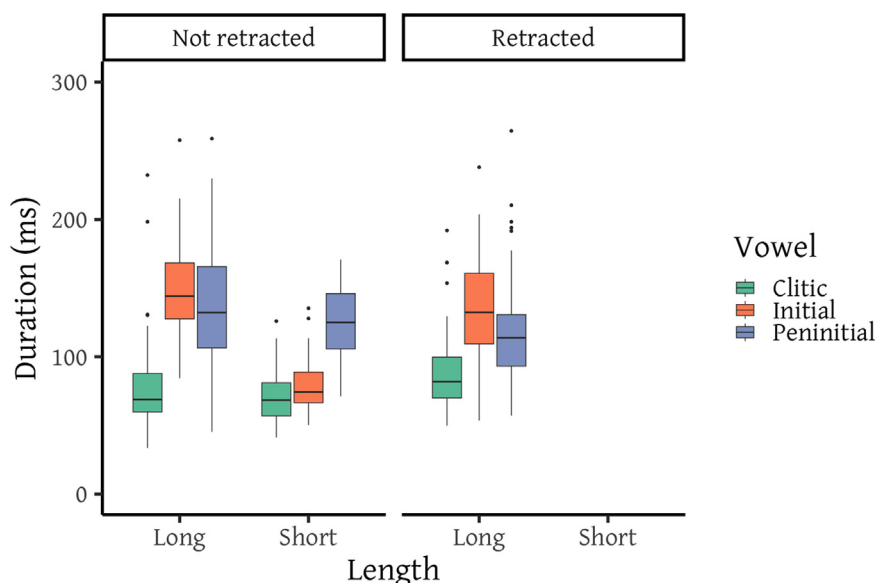


Fig. 7. Duration by the position of the vowel (clitic, initial, peninitial) in the elicited disyllables. Data is grouped by whether stress is not retracted (left two panels) or retracted (right two panels) and by phonological length of the word-initial vowel (Long, Short). The first three boxplots represent the three vowels in forms like [pɔ krááwɔ] 'after (the) cow' (16-b). Next are the vowels of forms with only short vowels, like [nɔ lípɔ] 'on (the) linden tree' (16-b) and finally, the vowels in forms with retraction such as [nó ʔááwɔ] 'on (top of the) head-ACC' (16-a). Retracted forms with only short vowels are unattested, see (16-a). Two tokens with the duration above 300 ms were excluded.

[nɔ 'lipo] 'on (the) linden tree' (16-b). We see that the duration of the clitic vowel is shorter than the initial, stressed long vowel. The peninitial vowel also has longer duration, but this has to do with phrase-final lengthening, a cross-linguistically common pattern (Oller, 1973; Klatt, 1975; Beckman & Edwards, 1990). Nearly all our elicitations had the relevant word at the end of a phrase because this is the most neutral word order for clitic contexts. Initial short vowels have shorter duration than corresponding long vowels, but the duration is slightly longer than the clitic vowel, which we attribute to the phonetic effect of stress: stressed vowels are longer than unstressed vowels of the same length. Moving on to the right-hand panel, we see words in the retracted condition, such as [nɔ 'ɣtaáwɔ] 'on (top of the) head-ACC' (16-a). The stressed, clitic vowel in the retracted forms (e.g. [nɔ 'ɣtaáwɔ]) has slightly longer duration when compared to matching non-retracted, unstressed clitic vowel ([pɔ 'kráɣwɔ]), which is the second boxplot in Fig. 7. However, all clitic vowels are substantially shorter than the following long vowels, be they stressed (as in non-retracted words) or unstressed (as in retracted words). This suggests that stress is phonetically realized as increased duration (e.g. short vowels are longer when stressed), but the phonological vowel length contrasts are maintained, with short vowels being shorter than long vowels, regardless of stress.

We move on to intensity in Fig. 8. We measured the mean intensity for the entire duration of the vowel. Here we see that peninitial vowels have the lowest mean intensity. While we use the term peninitial to refer to these vowels for consistency across figures, these vowels are also phrase-final and unstressed. Initial vowels in the non-retracted words (e.g. [pɔ 'kráɣwɔ] 'after (the) cow') have the highest intensity, since they are stressed. The clitic vowel has a slightly lower intensity in the non-retracted condition, but has the highest intensity in the retracted condition (e.g. [nɔ 'ɣtaáwɔ] 'on (top of the) head-ACC'). While these differences are small, the consistency across different groups suggests that mean intensity is a correlate of stress. When the clitic vowel is stressed, as in the

retracted condition, it will have a higher mean intensity than the following long, unstressed vowel.

The duration and intensity data so far suggest that stress is indeed realized on the clitic with retraction. How about the pitch? Fig. 9 presents normalized pitch tracks by vowel (clitic, initial, peninitial), depending on whether the forms show retraction or not. The initial vowel appears very similar to monosyllables (Fig. 2) and disyllables (Fig. 4), regardless of whether retracted or not. The pitch tracks converge on the peninitial vowel. As for the clitic vowel, the key distinction is the relative pitch height. In the non-retracted condition, the pitch on the clitic is crucially lower than the peak on the following vowel. While the three prosodic combinations (Long Falling, Long Rising, and Short) differ, they all contain a phonological High tone on the initial, stressed syllable. The relatively lower pitch on the clitic suggests no High tone on the clitic vowel. In contrast, the pitch in the retracted condition is higher (or equal) on the clitic when compared to the initial vowel for both Long Falling and Long Rising pitch. This is consistent with a High tone on the clitic. We can further see that the Long Falling and Long Rising words converge, with a peak at the end of the initial vowel, consistent with High tone on the second mora of that vowel. While there are small differences between the two, these can be attributed to phonetics and the fact that the segmental material has not been controlled for. What is clear is that the distinction between the two pitch accents is much smaller in the retracted condition, supporting a neutralization analysis. We interpret these data as having a High tone on the clitic and on the second mora of the initial, long vowel. Disyllabic stems with only short vowels never show retraction and are not shown.

We finally move to words with long [ɛɛ], which never show retraction. Fig. 10 presents pitch tracks for words with [ɛɛ]. As noted above, Long Rising pitch on [ɛɛ] is much more frequent than Long Falling (Stanonik, 1977 presents about ten times as many Long Rising as Long Falling words with [ɛɛ]). In our elicitations, we only had Long Rising disyllables with

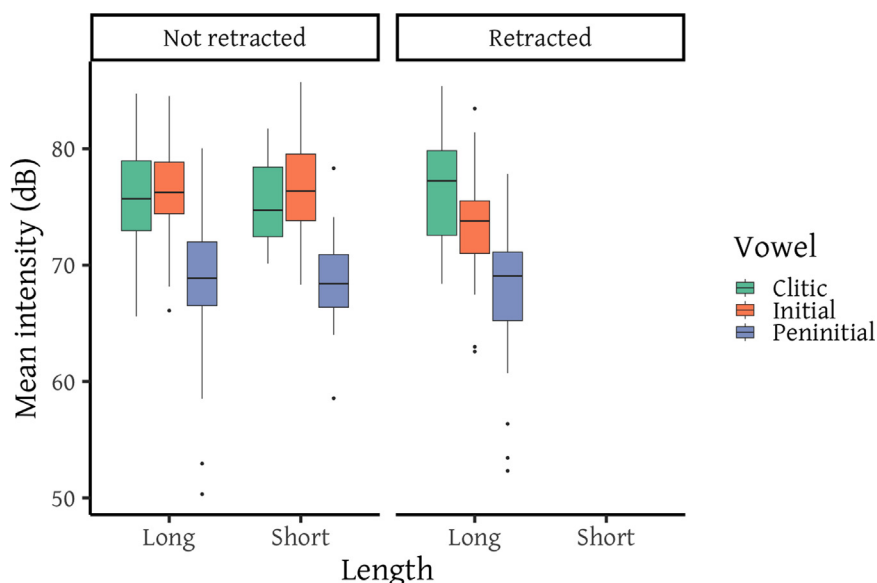


Fig. 8. Mean intensity in the elicited disyllables by the position of the vowel (clitic, initial, peninitial), retraction, and phonological length of the word-initial vowel (Long, Short). See Fig. 7 for example words.

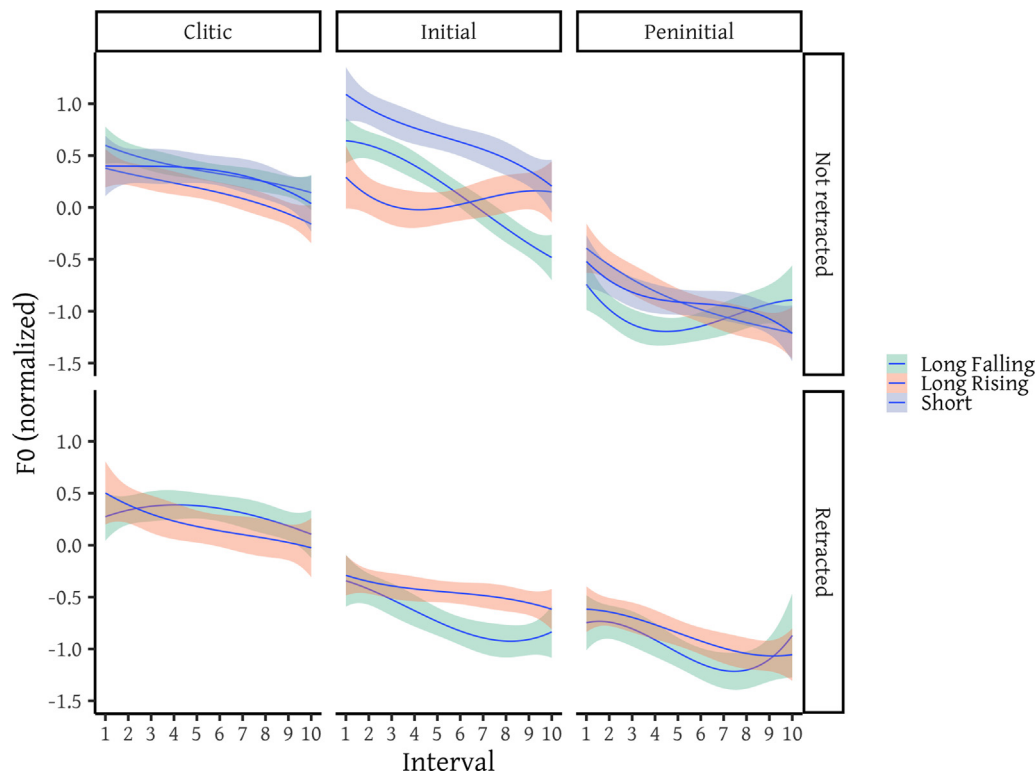


Fig. 9. Normalized pitch tracks in the elicited disyllables by the position of the vowel (clitic, initial, peninitial), retraction, and pitch (Long Falling, Long Rising, Short). At the top, the non-retracted forms include Long Falling [po 'króowo] 'after (the) cow' (16-b), Long Rising [no 'koóozo] 'on (the) goat' (16-b), and Short [no 'lipo] 'on (the) linden tree' (16-b). At the bottom, the retracted forms include Long Falling [nó ytoóowo] 'on (top of the) head-ACC' (16-a) and Long Rising [nó eézer] 'on (the) lake' (16-a). There are no forms with only short vowels and retraction, see (16-a).

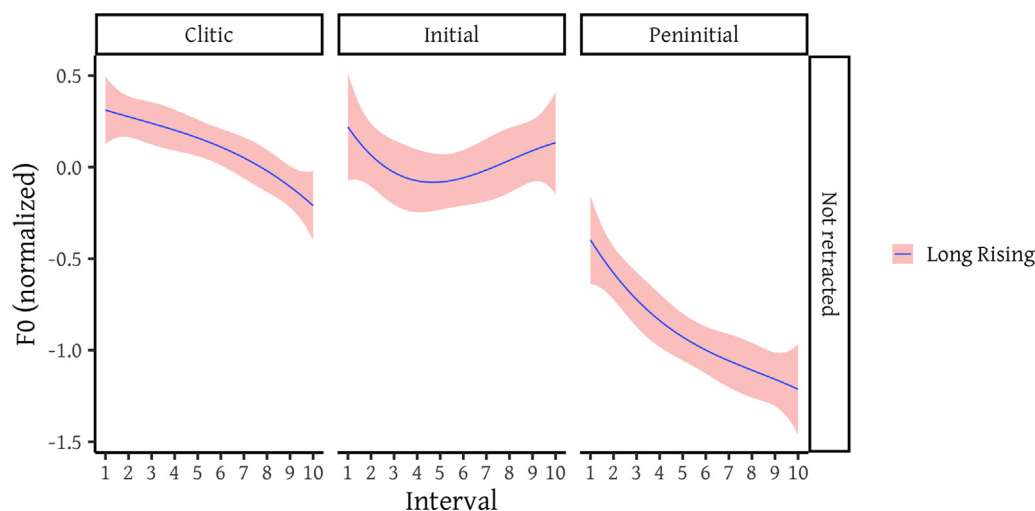


Fig. 10. Normalized pitch tracks for Long Rising disyllables with [εε], such as [u 'zeémle] 'in (the) soil, earth'.

[εε]. The data in Fig. 10 closely match non-retracted Long Rising words containing other vowels, as in Fig. 9. The acoustic analysis thus suggests that stress does not retract from [εε]. Phonologically, this means that no words with an [εε] allow for a floating High tone.

3.3. Interim summary

The data presented in this section allowed us to further explore the asymmetries first observed when looking at the

distributions (Section 2). The main finding is that clitic and content word combinations largely mirror distributions in non-clitic contexts: monosyllables with clitics strongly resemble disyllables without clitics, whereas disyllables with clitics strongly resemble trisyllables without clitics.

This generalization informs the phonological analysis. In Section 2 we compared disyllables with non-initial long vowels as having either one or two High tones, which results in either initial or final stress. The distributions tell us conclusively that long vowels must be linked to a High tone to distinguish

between rising and falling pitch. As for short vowels, their High tone does not need to be linked underlyingly.

The reasoning goes as follows. Disyllables with final long vowels distinguish two stress patterns for each pitch on the long vowel: Long Falling can have initial ([kókóof] ‘chicken’) or final stress ([kó'leen] ‘knee’), and the same is true for Long Rising ([ótruóp] ‘bran’ versus [p'traóts] ‘children’). The distinction between initial and final stress depends on the number of High tones: all forms must have a linked High tone on the long vowel, and the additional High tone will shift the stress one syllable to the left, whether it is linked or floating. The floating tone analysis shown in (15) works for disyllables with final long vowels just as it does for monosyllables with long vowels preceded by a clitic. The floating tone must first link to the short vowel after which it becomes stressed because it is leftmost in the prosodic word. The floating tone analysis can be extended to words with only short vowels, where floating tones distinguish between retracting and non-retracting monosyllables.

This analysis also correctly predicts the patterns in disyllables with initial long vowels. When these have a floating tone, it will link to the preceding clitic if one exists, shifting stress to the left. The additional restriction is observed when the two High tones would fall on adjacent moras: this is why falling [ʔááwə] ‘head-ACC’ becomes rising [nó ʔááwə] ‘on (top of the) head-ACC’ when preceded by a clitic.

Comparing phrases of clitics and content words, we can make several observations. First, monosyllables display a greater range of patterns than disyllables. Among content words with only short vowels, monosyllables may exhibit retraction, whereas disyllables never do. While Long Falling and Long Rising monosyllables remain distinct in monosyllables with retraction, Long Falling disyllables become Long Rising. Second, there is a key vowel quality difference. While most vowels can appear in retracted and non-retracted stems, long [ɛɛ] effectively blocks retraction, regardless of whether falling or rising, or whether appearing in monosyllables or disyllables. This means that words with long [ɛɛ] do not have a floating High tone. This restriction does not apply to short [ɛ]. It is unclear what the motivation for these patterns is. In the following section we thus explore the productivity of these asymmetries. In particular, we ask whether native speakers mirror the difference between monosyllables and disyllables, between Long Falling and Long Rising disyllables, and between [ɛɛ] and other vowels.

4. Tonal wug-test experiment

Section 2 established the main generalizations about the distribution of vowel length, tone, and stress in Žiri. In Section 3 we have seen that Žiri has a complex prosodic system that shows a key prosodic alternation—retraction. While retraction mirrors the prosodic distribution found in isolated words, it also allowed us to further probe the prosodic restrictions. In particular, we found that (i) monosyllabic words are less restricted than disyllabic words, (ii) Long Falling disyllables become rising when retracted, and (iii) vowel quality is a key factor. However, our generalizations are based on a limited number of elicitations, which hinders the reliability of our claims. In this section, we tackle the question whether these asymmetries are productive using a nonce-word experiment.

We follow a growing body of literature that uses nonce-word experiments to corroborate speakers’ grammatical knowledge (Berko, 1958). For instance, Ernestus and Baayen (2003) show that Dutch speakers’ productions of nonce words reflect distributional characteristics of the Dutch lexicon, with velars in root-final position eliciting relatively more voicing than labials and coronals. For Slovenian, Jurgec and Schertz (2020) show that velar palatalization at the root-suffix boundary is disfavoured when the stem contains another postalveolar due to a consonant co-occurrence restriction confirmed on corpus data (Jurgec, 2016).

Even though the tonal generalizations are quite robust in the lexicon, research has shown that not all trends in the lexicon are productively extended to nonce words. In Becker, Ketrez, and Nevins (2011), for instance, the Turkish lexicon displays different rates of laryngeal alternation in stops depending on vowel quality, but this was not mirrored in the nonce word task, although many other generalizations were. As for tone, Zhang et al. (2006) show that while several tone-sandhi alternations are completely regular in Taiwanese, they are not extended to nonce words. In Zhang and Lai (2010), participants extended a phonetically motivated Mandarin tone-sandhi to novel words; another type of sandhi, which lacks clear phonetic motivation, however, was applied at significantly lower rates.

The productivity is a key issue for Žiri, where [ɛɛ] inhibits stress retraction. While there are some examples of interactions of prosody with vowel quality (see Becker & Jurgec, 2017; Becker & Jurgec, 2020 for a review) or sonority (de Lacy, 2006; de Lacy, 2007), the Žiri type of interactions do not have clear phonetic grounding. In particular, while the intrinsic pitch of vowels is dependent on vowel height, which could lead to tonogenesis, this analysis is not possible for Žiri. In order for this analysis to work, mid vowels would have to pattern with high or low vowels with respect to the distribution of tone. This is not what we find in Žiri: the mid vowel [ɛɛ] patterns differently than high and low vowels, as well as the two other mid vowels [ee] and [oo] and diphthongs. The same reasoning applies to sonority, and to the relationship between sonority and stress.

The restriction on vowel quality is only one issue we explore in the experiment. The examination of distributions and alternations also revealed two other patterns where we expect asymmetries. First, the patterns in monosyllables differ from those observed in disyllables. Long Falling monosyllables show retraction while retaining the falling contour, whereas the disyllables show a tonal change on the long vowel, resulting in neutralization with Long Rising disyllables. In the experiment, we ask whether participants treat Long Falling monosyllables differently than disyllables. Given that the contour is preserved faithfully in the retracted monosyllables but not disyllables, we would expect participants to prefer more retraction with monosyllables. Put differently, all else being equal, participants would prefer uniform paradigms, which is a well-established effect in phonology (Benua, 1997; Kenstowicz, 2004; Burzio, 2005; Steriade, 1997).

The second asymmetry concerns the two types of disyllables. As we have seen above, Long Falling disyllables become Long Rising when retracted. Long Rising disyllables, on the other hand, retain the Long Rising contour. If participants were to distinguish between the two, we would expect a greater

preference for retraction with Long Rising pitch when compared to Long Falling. This is again related to the preference for uniform paradigms.

These three generalizations can be formalized as hypotheses (17).

(17) Hypotheses

- a. Word-length: Retraction in Long Falling monosyllables is preferred over retraction in Long Falling disyllables.
- b. Tone: Retraction in Long Rising disyllables is preferred over retraction in Long Falling disyllables.
- c. Vowel quality: Retraction from [ɛɛ] is dispreferred compared to other vowels.

To assess these hypotheses, we conduct a perceptual wug-test. Participants were asked to judge nonce-word paradigms. As we will see, the results confirm all three hypotheses. We lay out specific predictions in the following section.

4.1. Methods

Materials. Our stimuli consisted of phonotactically valid words that were generated to vary on several parameters (summarized in Table 1). The monosyllables were of the shape CVC whereas the disyllables were 'CVCb. The words could have either Long Falling, Long Rising, or Short pitch accent. However, not all combinations of tone and word length were included. For instance, there were no Long Rising monosyllables, as those are much less common in the lexicon than Long Falling monosyllables (see Section 2.3). We also did not include disyllables with only short vowels, as they never show retraction. In short, the gaps in the stimuli reflect the gaps in the lexicon, yet we will still be able to answer the key question about the interaction of stress, tone, quantity, and vowel quality. The final variable was vowel quality: among the Long Rising disyllables, some had the vowel [ɛɛ]. All other long stressed vowels were [ee, aa, oo], whereas the short vowels were [i, ɛ, o]. About two thirds of the stimuli were minimal *n*-tuples, as the disyllabic pair in Table 1, while the remaining stimuli were not paired. We provide more detail when discussing individual comparisons in the section below. The stimuli were checked with the native speakers to make sure they were not real words.

These stimuli were chosen to test the hypotheses in (17). Regarding word length (17-a), we predict that for monosyllables like ['xót] participants will prefer the retracted ['nó xót] (when also offered the non-retracted [no 'xót]) more often than for Long Falling disyllables like ['réelo] and the retracted form ['nó réelo] (when also offered the non-retracted [no 'réelo]).

With respect to tone, we predict that for Long Rising disyllables like ['réelo] participants will prefer the retracted ['nó réelo] (when also offered the non-retracted [no 'réelo]) more often than for Long Falling disyllables like ['báamo] and the retracted form ['nó báamo] (when also offered the non-retracted [no 'báamo]). Finally, regarding vowel quality, we predict that for words with [ɛɛ] like ['réelo] participants will prefer the retracted ['nó réelo] (when also offered the non-retracted [no 'réelo]) less often than for words with other vowels like ['báamo] and the retracted form ['nó báamo] (when also offered the non-retracted [no 'báamo]).

In total, there were 82 words, which are broken down by group in Table 1; see Appendix for a full list. For each stem, we recorded a triplet: a bare form and two forms with a preposition, one with retracted stress and the other without. Stimuli were recorded in an anechoic room with a Sound Devices USB Pre 2 pre-amplifier with 44.1 kHz sampling using AKG C544L cardioid condenser head-mounted microphone. We RMS-equalized stimuli in Praat (Boersma & Weenink, 1992, using a Praat script by Beckers, 2002) and converted .wav to .mp3 format with Audacity's LAME mp3 encoder for faster processing. The visual stimuli of unusual objects or non-objects were taken from Horst and Hout (2016) and Gonzales-Gomez, Poltrock, and Nazzi (2013).

Eliciting stimuli with tonal trajectories as complex as those in the Žiri poses unique challenges. To achieve the desired tonal trajectories of nonce words, the researcher (one of the authors) produced tonal trajectories and then asked a native Žiri dialect female speaker (age 26) to repeat them. This methodology was necessary due to the speaker's difficulty in producing tonal trajectories in nonce words, and the task was facilitated by the researcher's metalinguistic awareness and native competence in the relevant dialect. Each item was recorded several times in two separate sessions. The final triplets (stimulus and two options) were chosen after a careful acoustic analysis in Praat to make sure that the trajectories matched the patterns found in natural speech. Altogether 246 items were chosen (82×3 for each triplet).

Procedure. The experiment was conducted online using Experigen (Becker & Levine, 2012). IRB approval was obtained, and speakers were asked to complete an online consent form. Participants were recruited for the experiment with the help of a local radio station, personal connections, and a paid social media post that targeted inhabitants of the Žiri Valley. At the beginning of the experiment, the participants were asked to volunteer their demographic information, including age and place of residence. Participants were then told that they will hear three words. The first one will be a name of an imaginary object. The participants would then hear two

Table 1
Summary of parameters for nonce word generation. The numbers represent the number of stimuli.

Variables			Examples		
Words	Pitch accent	Vowel quality	Without clitic	Non-retracted	Retracted
Disyllables (63)	Long Falling (27)	other	'réelo	no 'réelo	'nó réelo
	Long Rising (36)	[ɛɛ] (12)	'réelo	no 'réelo	'nó réelo
Monosyllables (19)	Long Falling (11)	other (24)	'báamo	no 'báamo	'nó báamo
		other	móos	no 'móos	'nó móos
		other	'xót	no 'xót	'nó xót

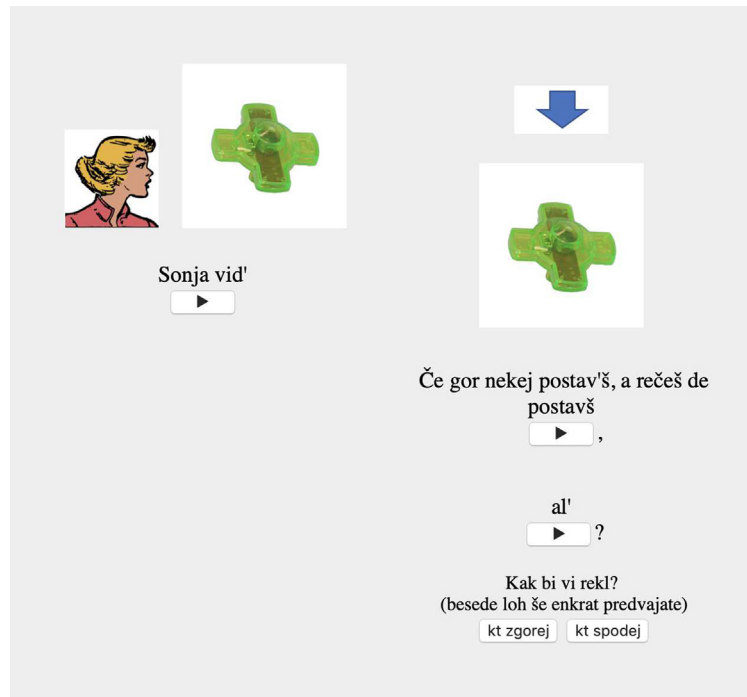


Fig. 11. Web interface for the nonce word study using Experigen (Becker & Levine, 2012) and visual stimuli (non-objects) from Gonzales-Gomez et al. (2013) and Horst and Hout (2016).

versions of the same object in the clitic context. Each trial started with the picture on the left in Fig. 11, under which the frame sentence “Sonya sees (button)” was written. When the participants clicked on the button, they heard the stimulus in the accusative and without the preposition. Then an additional picture appeared with an arrow above the object and the sentence “If you put it on something, would you say (button) or (button)?” When selected, the buttons played the stimulus with the preposition [no] ‘on’. This preposition was chosen because it does not change the case of the bare form (both accusative). One of the stimuli had retracted stress and the other had the stress on the stem. Individual stimuli had constant placement, but the order of trials was randomized for each participant. Retracted and unretracted words were approximately equally assigned to two positions across different groups of words. Adding a random intercept for the stimuli order does not change any significance levels of the estimates, which suggests the order of the stimuli does not interfere with experimental results.

Once the stimuli were played, the sentence “What would you say? (You can play the words again.)” appeared as well as two buttons: “like above” or “like below”. The participants were free to listen to the materials as many times as they wanted. Once they made a judgment, the next stimulus was played. All text was written using informal conventions (e.g. *vid* instead of the standard *vidi* ‘sees’, *rekl* instead of the standard *rekli* ‘said’) to prime the participants for non-standard stimuli. Stimuli in the current experiment were never presented orthographically.

Participants. We include in the analysis only those participants who indicated on the demographic survey that they were native speakers of the Žiri dialect without any linguistic training. 82 such participants responded to at least one judgment, but

we include only participants who provided more than 10 judgments in the final analysis. Participants with fewer responses were excluded from the analysis in order to avoid participants who were just curious about the experiment and did not commit to listening to at least a few stimuli. We ran all our final models on data sets with all participants as well as only those participants who made all 82 judgments; estimates of interest are significant in all three data sets with very similar estimates. Altogether 56 participants are included in the final analysis. Their self-reported mean age was 32 (the range was between 19 and 57 with a median of 29 years). 24 participants provided judgments to all 82 stimuli. The mean number of judgments for the remaining participants is 24 (ranging from 11 to 74, with a median of 18).

4.2. Results

The aim of the experiment was to explore whether word length, tone, and vowel quality have an effect on retraction. We find that the participants preferred retraction with monosyllables and with Rising pitch, but dispreferred retraction from [εε]. We review these results in turn.

Word length. To test if rates of retraction differ depending on word length, we analyze retraction rates from 11 pairs (22 items) of monosyllabic and disyllabic words with the Long Falling pattern. These minimal pairs differ only in the number of syllables, but are otherwise segmentally and suprasegmentally identical. For example, for the monosyllabic form [xáat], participants chose between the non-retracted [no xáat] and retracted [nó xáat]. For the non-clitic form [xáato], the participants chose between the non-retracted [no xáato] and retracted [nó xoáto]. We also include 16 disyllabic words with the Long Falling tone without the monosyllabic pair. We include

these additional items because the random slopes in our statistical analysis reveal that segmental material does not influence experimental outputs; we elaborate on these below.

We fit the data into a mixed-effects logistic regression model with Response as the dependent variable (retraction coded as success; $N = 1259$) and two predictors: Word Length and Vowel Quality ([aa, ee, oo]; sum-coded with [oo] as the reference level). The full model includes an interaction between the two predictors as well as random intercepts for participant and frame with by-participant and by-frame random slopes for Word Length.

To control for effects of segmental material and neighborhood density, we include random intercepts and slopes not by item, but by *frame*. For example, [xáat] and [xáatə] are coded as one frame, because they involve the same segmental material, except that [xáatə] includes an additional syllable—but the number of syllables is a fixed effect and as such already controlled for in the model. Items that are not part of the mono- and disyllabic pairs are coded as unique frames. We model random intercepts for frames rather than items, because this allows random slopes for the predictor of interest (see also Beguš, 2017 for a similar approach). Had we instead chosen items and random slopes for the predictors of interest, the models do not converge as each item has only one level of the predictor of interest.

The interaction between Word Length and Vowel Quality is not significant. In particular, the Akaike Information Criterion (AIC), a measure of the quality of statistical models, is 1385.6 in the model without the interaction and 1387.8 with it. Word Length is a significant predictor (AIC is 1385.6 in the model with the predictor and 1399.5 in the model without the predictor). The estimate of the difference between disyllables and monosyllables (β) is 1.24, while the z-score is 5.04, which is statistically significant ($p < 0.0001$). Fig. 12 illustrates the effects with 95% confidence intervals.

By-participant and by-frame random slopes reveal relatively little variability across participants and frames. Random slope

for the difference between disyllables and monosyllables is positive for all frames and for all participants. This also suggests that segmental material and neighborhood density do not crucially affect the results of the experiment.

Tone. To test if retraction preference differs between Long Falling and Long Rising tones, we analyze the data from 46 disyllables. The first vowel in the disyllables was either [ee] or [aa]. The data included 17 minimal pairs (34 items in total) in which the segmental material was constant, the only difference between them was the tonal trajectory of the stimulus (Long Falling versus Long Rising). For example, for the Long-Falling non-clitic form [báámə], speakers chose between the non-retracted [nə 'báámə] and retracted [nó báámə]. On the other hand, for the Long-Rising non-clitic form [baámə], speakers chose between the non-retracted [nə 'baámə] and retracted [nó baámə]. In addition to 34 such minimal pairs, the data also includes 12 items in which the Long Falling and Long Rising patterns were not segmentally identical (5 are Long Falling, while 7 are Long Rising).

We fit the data into a mixed-effects logistic regression model with Response as the dependent variable (retraction coded as success; $N = 1516$) and two predictors: Tone (Long Falling and Long Rising; treatment-coded with Long Falling as the reference level) and Vowel Quality ([aa] vs. [ee], sum-coded with [ee] as reference). The full model includes an interaction between the two predictors as well as random intercepts for participant and frame with a by-participant and by-frame random slopes for Tone. The interaction is not significant (AIC is 1701.5 in the model with the interaction and 1699.9 without it). Tone is a significant predictor (AIC is 1699.9 in the model with the predictor Tone and 1710.9 in the model without it). The estimate is 0.90, with the z-score of 3.914, which is statistically significant $p < 0.0001$. Due to convergence, the model without predictor Tone had to be fit with the Nelder-Mead optimizer (all other models are estimated with the BOBYQA optimizer). Fig. 13 illustrates the effects with 95% confidence intervals.

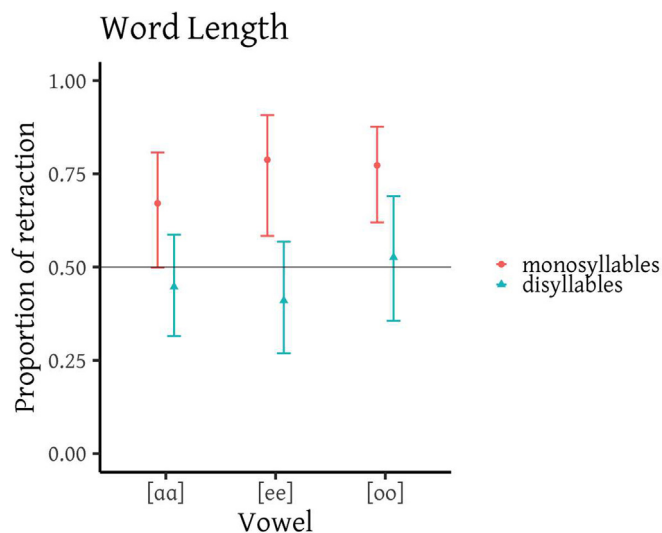


Fig. 12. Estimates of a mixed-effects logistic regression model showing preference for retraction in monosyllables with long vowels other than [ee] in the Falling tone condition. Error bars indicate 95% confidence intervals.

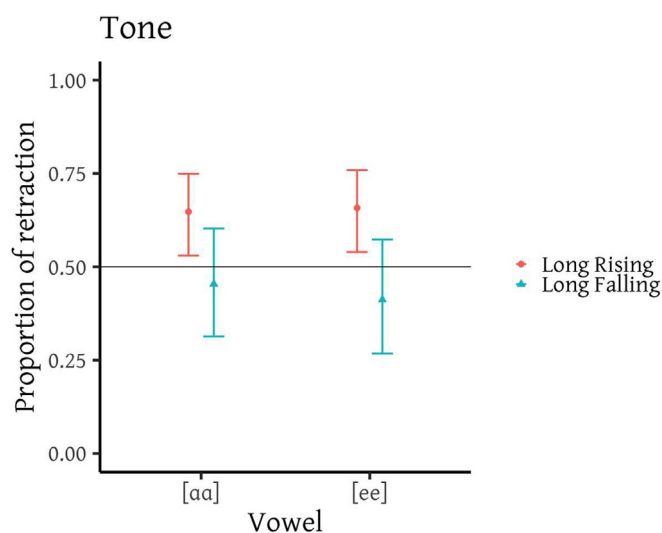


Fig. 13. Estimates of a mixed-effects logistic regression model showing preference for retraction for Long Rising (compared to Long Falling) with long vowels other than [ee]. Error bars indicate 95% confidence intervals.

Estimates of the random slopes for the difference between Long Rising and Long Falling are positive for all frames. The same estimates are negative for only three out of 56 subjects. This again suggests that segmental material or neighborhood density do not crucially affect the results and that subjects are highly uniform in their responses.

Vowel quality. To test the effect of vowel quality on retraction, we analyze disyllables with the Long Rising pattern and vowels [ee, aa, εε] to maximally control for other influences. For example, for the non-clitic form ['bέεmp], participants chose between the non-retracted [no 'bέεmp] and retracted ['nó bέεmp]. For ['béεmp], the participants chose between the non-retracted [no 'béεmp] and retracted ['nó béεmp].

The items include 7 minimal triplets (a total 21 items) that differ only in the first vowel ['bέεmp, 'béεmp, 'báámp]. The two items with mid vowels are coded as one frame; the item with [aa] is coded as a separate frame. Additionally, we analyze five item-pairs (10 in total) that differ only in [ee] and [εε] as one frame each. Finally, we include five items without pairs that include Long Rising tone and vowel [aa] (separate frames) for a total of 36 items (with corresponding retracted and non-retracted options).

To test if rates of retraction differ between the three vowels, we fit the data to a mixed-effects logistic regression model with Response as the dependent variable (retraction coded as success, $N = 1201$) and the predictor Vowel Quality ([ee, aa, εε]; treatment-coded with [εε] as the reference level). The full model includes random intercepts for participant and frame with by-participant and by-frame random slopes for Vowel Quality. As Table 2 shows, retraction is significantly less frequent than chance for vowel [εε]. Retraction is also significantly more frequent for vowel [ee] when compared to [εε] and more frequent for vowel [aa] when compared to [εε]. Fig. 14 illustrates the effects with 95% confidence intervals. No overdispersion was detected in the three models: estimates are 0.82, 0.83, and 0.77 (using the `overdisp_fun()` from Bolker, 2022).

Again, random intercepts and slopes suggest that participants are highly uniform in their responses. All estimates for by-frame random intercepts and slopes conform to the estimates of the main effects with no deviations. The difference between [εε] and [ee] is not positive for only three participants (out of 56), and the same is true for the difference between [εε] and [aa] for one participant.

To summarize, the nonce-word experiment results suggest that retraction is preferred in monosyllables when compared to disyllables and in words with Rising pitch when compared to Falling pitch. Finally, [εε] inhibits retraction when compared to other vowels.

Table 2

Estimates of a mixed-effects logistic regression model showing preference for retraction from [ee] and [aa] compared to [εε] (Intercept) in disyllabic words with the Long Rising pattern. SE stands for Standard Error.

	β	SE	z	p
(Intercept)	-1.006	0.265	-3.802	0.0001
Vowel [ee] (versus [εε])	1.740	0.305	5.698	0.0000
Vowel [aa] (versus [εε])	1.739	0.285	6.108	0.0000

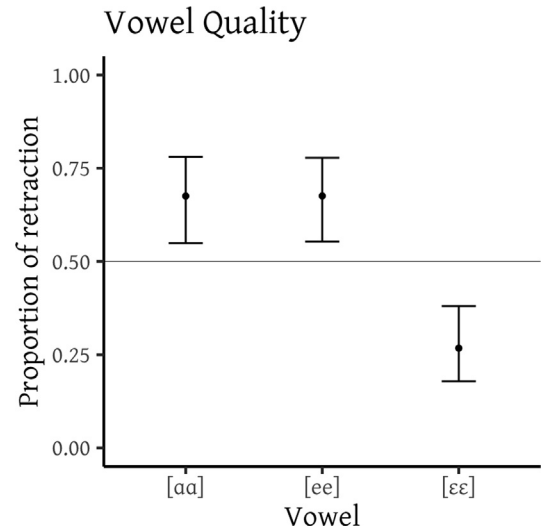


Fig. 14. Estimates of a mixed-effects logistic regression model showing preference for retraction from [ee] and [aa] compared to [εε] in disyllabic words with the Long Rising pattern. Error bars indicate 95% confidence intervals.

4.3. Discussion

The experiment allowed us to tackle the questions that go beyond the analysis of the data reported in Sections 2 and 3. In particular, we confirmed that retraction is preferred in monosyllables when compared to disyllables, and in words with Rising tone when compared to Falling tone. Participants also strongly dispreferred retraction from [εε]. In this section, we offer explanations for these patterns.

Word length. Participants preferred retraction in monosyllables at rates significantly higher than chance levels and in disyllables. This result is not surprising, since monosyllables often behave differently than disyllables. For instance, while BCMS contrasts rising and falling tones in disyllables and longer words, monosyllables allow only falling tone. Rising tone requires a posttonic syllable, which is only possible in disyllables (Inkelas & Zec, 1988; Zec, 1999; Langston, 1997). When it comes to segmental patterns, monosyllables are entirely protected by initial syllable faithfulness, but disyllables or longer words are not (Beckman, 1997). In Tamil, coda nasals of the stem assimilate to the onset of the suffix, except in monosyllabic stems (Christdas, 1988).

In Žiri, however, it seems that stress retraction in monosyllables has a simpler explanation, which has to do with a preference for retaining the same tonal pattern. In the experiment, we were comparing monosyllabic and disyllabic Long Falling words. Recall that in disyllables with the Falling tone there is an alternation in the long vowel: the non-clitic form ['báámp] retains its falling contour when preceded by an unstressed clitic, but changes the contour to rising when preceded by a stressed clitic: ['nó báámp]. No such alternation is found in monosyllables: falling ['páám] is maintained in ['nó páám]. Hence the participants might prefer no tonal changes, and thus disfavour retraction in disyllables which necessarily leads to a tonal alternation on the long vowel. This reveals that an output-to-output faithfulness constraint on tonal identity inhibits stress retraction in disyllables, but has no effect on monosyllables. Under this view, retraction is a function of word length only indirectly: participants prefer to retain the contour on the long

vowel, which applies equally to all words. Disyllables with stress retraction neutralize tone on the posttonic vowel while monosyllables do not. These results are consistent with other experimental findings. All other things being equal, participants will prefer uniform paradigms. There is a variety of theoretical approaches that capture this concept, such as Output-to-Output Correspondence (Benua, 1997), Paradigm Uniformity (Kenstowicz, 2004; Burzio, 2005), and Lexical Conservatism (Steriade, 1997).

One could ask whether the productive generalizations are made at the level of the underlying representations or surface forms. While most phonological research has focused on output-based generalizations (Ernestus & Baayen, 2003; Hayes, Zuraw, Siptár, & Londe, 2009; Jurgec & Schertz, 2020), there is also evidence about input-based generalizations. For instance, Becker and Gouskova (2016) show that Russian speakers make generalizations about mid vowel deletion both based on the input as well as output. While consistent with the Russian analysis, the Žiri data are perhaps better interpreted through paradigmatic relationships: the participants evaluated paradigms, generally preferring uniformity, except when this is not possible, as in Long Falling disyllables.

Tone. We also saw that rising tone prefers more retraction than falling tone. In non-retracted configurations both types of tones are distinctive. In retracted configurations, however, the contrast is neutralized: only rising tone is possible on the surface. For instance, [nó baómp] can be the clitic form of [baómp] or [báomp]. The tonal change in the words with falling tone can be explained as an Obligatory Contour Principle (OCP) effect: two adjacent tones are a marked situation, which is resolved by shifting the High tone from the first mora to the second one when preceded by a High tone on the clitic. Cross-linguistically, the prohibition on adjacent identical tones may be resolved by deleting one of the tones or shifting one of the tones so that they are no longer adjacent (Myers, 1997). The latter situation is what happens in Žiri disyllables.

Returning to the asymmetry between the Rising and Falling tones, Output-to-Output Correspondence becomes relevant yet again. Participants preferred more retraction with Rising tones, which retain the same contour regardless of whether they are retracted or not. Long Falling disyllables prefer less retraction than Long Rising disyllables. This is because a retracted form would lead to tonal change, violating the pressure for the uniformity of tone across all instances of the same word. This complements the results seen in monosyllables, where we also saw a significant preference for uniform paradigms.

Vowel quality. We finally move to vowel quality which also interacts with retraction. This interaction is tied to a specific vowel quality: retraction from [ɛɛ] is dispreferred compared to the other vowels. This type of interaction between segmental and prosodic properties is very unusual. Most other types involve sonority, but in Žiri this is clearly not the case. High sonority vowels may prefer stress (Kenstowicz, 1997), but [ɛɛ] is intermediate in sonority between [ee] and [ɑɑ] which we had in our nonce words. de Lacy (2006) and Blumenfeld (2006) make an explicit claim that sonority, but not segmental features, can interact with stress assignment and tone, and Rasin (2017), Shih (2016), Shih (2018), Shih and de Lacy (2020) argue that even sonority cannot affect stress placement. Our results offer clear evidence that vowel quality interacts with prosody, and that this is unre-

lated to sonority. Retraction from [ɛɛ] is dispreferred in production as well. The speaker who recorded the stimuli had difficulties producing the retracted pattern with this vowel.

A similar pattern is found in Standard Slovenian. There are both tonal and stress restrictions. Lax mid vowels cannot have a falling tone in Standard Slovenian (Becker & Jurgec, 2017): in native words, this means that the tone will change to accommodate the vowel quality, whereas in loanwords, which generally only have falling tone, the vowel tenses. There is a second interaction between vowel quality and stress: lax vowels are marked and avoid stress. Becker and Jurgec (2020) analyze this situation in terms of positional faithfulness, which is satisfied by shifting stress to the following vowel in disyllables. When stress is stem-final, the vowel quality changes instead, and monosyllables are faithful.

Žiri offers another piece of evidence that interactions between vowel quality and prosody are attested. When compared to Standard Slovenian, however, there are a few key differences. First, it is only [ɛɛ] that is affected, since there is no corresponding back mid lax vowel. Second, [ɛɛ] in Žiri cannot be posttonic, regardless of its tone. When stressed, [ɛɛ] can have a Rising or less commonly, Falling tone. When stress retracts from the Falling tone, the resulting tone is Rising in non-final syllables, and Falling in the word-final syllable. Because retraction is strongly disfavored, both tonal patterns are ruled out on unstressed [ɛɛ] and both are possible on stressed [ɛɛ], so the restriction in Žiri cannot be related to tone alone, but has to do also with footing. One way to capture that is to say that [ɛɛ] has to be aligned with the left edge of the foot. However, because stress and footing is predictable from High tone, we also need to conclude that a floating High tone is not possible in words with [ɛɛ] or at the very least, a floating High tone cannot dock to the preceding short vowel.

This type of prosodic restriction complements the two other patterns in Standard Slovenian: a markedness constraint on a combination of tone and vowel quality (Becker & Jurgec, 2017) or a positional faithfulness constraint on a specific prosodic position and vowel quality (Becker & Jurgec, 2020). In Žiri the positional faithfulness analysis is not available, because stress can shift (from all vowels but [ɛɛ]) to another instance of [ɛ]. Moreover, the short [ɛ] does not show the same restrictions as long [ɛɛ]. In the elicitation data, we have both retracted (13) and non-retracted monosyllables with [ɛ] (14). In the experiment, the short [ɛ] nonce words showed similar, above-chance rates of retraction to other short vowels (see Appendix).

There are historical reasons why [ɛɛ] behaves differently than other vowels. The prosodic patterns of Slovenian are largely determined by various retractions that occurred throughout history (Ramovš, 1928; Ramovš, 1936; Ramovš, 1950; Rigler, 1970; Greenberg, 2000). Most retractions involved only a subset of pitch accents and word positions. These retractions can be sensitive to vowel quality. In Standard Slovenian, one of the retractions applied only when the final falling/high ('circumflex' in the Slavic literature) pitch retracted to the preceding mid vowels and became rising/low ('acute'): *žɛ'ná* > *žěna* 'woman, wife'.⁷ The newly stressed short vowel subsequently lengthened, creating a new type of pitch accent on penultimate lax mid vowels.

⁷ To make this discussion more accessible to non-specialists, we use the IPA tonal notation rather than the one common in the Slavic linguistic literature.

Žiri appears to exhibit the same type of retraction, although Beguš (2011) suggests it is more recent than in other varieties of Slovenian. Žiri has several prosodic changes that precede retraction to mid vowels and are not shared by other varieties of Slovenian. Historically, retraction to the clitic affects all vowels and has to precede retraction to lax vowels. Like in Standard Slovenian, the retraction can be seen as High tone retracting one mora to the left, resulting in the High tone on the preceding short vowel, which could be within the stem or on the preceding clitic. The key difference is the position within the word. In disyllables or longer words with a final long vowel, the pitch was maintained on the long vowel (*večfjer* > *ʋečfjer* 'evening'), whereas in non-final syllables it subsequently changed to rising (*no yáawo* > **nó yáawo* > *nó yáawo* 'on (the) head'; *i'mieno* > **imieno* > *imieno* 'name.PL'). This retraction was accompanied by metatonic processes that changed Rising Long pitch into falling and vice versa in the words that did not have retracted stress (*kraáwo* > *kráawo* 'cow', *soóntse* > *soóntse* 'sun').

At this stage, long mid lax vowels did not exist in Žiri but were created by the subsequent retraction. Unlike other processes, this retraction was specific to mid vowels: tone shifted one mora to the left, retracting stress and concurrently lengthening the vowel (*méduét* > *méduet* 'bear'). The mid vowel retraction created Long Rising lax mid vowels, which did not exist in Žiri before; this emergence of a new vowel-dependent prosodic combination resembles the situation in Standard Slovenian reviewed above. The retraction also explains why Long Rising [é] is much more common than Long Falling.

However, there are two key differences between Standard Slovenian and Žiri, which have to do with the asymmetry between the front and back lax mid vowels and the tonal contrast on lax mid vowels. Only in Žiri do front mid vowels differ from back vowels. This is because the back mid vowel already merged with the low vowel before retraction, so the back vowels do not show the restrictions found in [é]. As for the tonal contrast, some vowels subsequently deleted and shifted High tone one mora to the left, resulting in Long Falling [é] (*sto'déénets* > *sto'déents* 'stream') in Žiri; no equivalent tonal change is found in Standard Slovenian.

These historical developments explain why [é] differs from all other Žiri vowels: retraction to the clitic preceded emergence of Long Rising [é], and further retractions due to deletion only shifted the High tone one mora (but not syllable) to the left, leading to a small number of Long Falling [é]. Back mid lax vowels merged with low vowels, which is why we do not see the same patterns with back vowels.

Our experiment shows that participants productively extend the restriction on [é] to nonce words. There is no obvious phonetic grounding for the pattern, regardless if the interaction is about stress in combination with quantity, as argued above, or tone. Phonetically unmotivated processes are fairly well understood at the segmental level (Blevins, 2004; Kiparsky, 2006; Kiparsky, 2008; Coetzee & Pretorius, 2010; Beguš, 2018; Beguš, 2019; Beguš, 2020). For instance, in Anishinaabemowin (Ojibwe) [n] alternates with [j]: [ki-na:n-a:] 'you fetch him' versus [ki-na:-im-i] 'you fetch us' (Buckley, 2000). There is no apparent phonetic motivation for this process, but it is not obviously antagonistic to some universal phonetic tendency (Beguš, 2019). Žiri provides an example of a

phonetically unmotivated process that involves prosody and a segmental feature. The results of the wug-test clearly document such an interaction in Žiri, complementing the data in other varieties of Slovenian.

5. Conclusions

This paper provides a description of complex prosodic interactions in Žiri Slovenian. We have seen that Žiri displays an interaction of tone, stress and quantity, which further interact with vowel quality. Stress in Žiri docks to the leftmost syllable with a High tone. Stress falls on the long vowel or the syllable immediately preceding it. In words with only short vowels, stress is penultimate. High tone can only surface within the foot.

The complexity of the system defines a new frontier of what a pitch accent language might look like. Although pitch accent as a term is controversial (see Hyman, 2006; Hyman, 2009; Hyman & Leben, 2021), it has been used to characterize prosodic patterns in languages ranging from Swedish, Latvian, Japanese to Basque, Luganda, and Yaqui. Compared to these languages, Žiri displays a greater number of prosodic contrasts. Disyllables, for instance, fully contrast seven different prosodic shapes based on the combination of stress, tone, and quantity. All prosodic contrasts fall within a single disyllabic foot, even in trisyllabic or longer words, contrasting Žiri from tonal languages like Mandarin, Vietnamese or Mazatec.

Žiri displays active alternations in which stress retracts to the preceding clitic. These alternations display several preferences, the productivity of which we corroborate in a perception experiment. We find that the participants generally prefer uniform paradigms. This means that there is more retraction in monosyllables when compared to disyllables because only the latter involve a tone alternation. Moreover, participants prefer more retraction from rising tone than from falling tone. This again has to do with the fact that retraction from a falling tone requires a tonal change, but this is not the case for rising tone paradigms in Žiri.

The key finding of this paper concerns interactions of prosody and vowel quality. The particular restriction in Žiri involves the long mid lax vowel [é], which displays lower rates of retraction when compared to tense mid vowels and low vowels. This interaction is thus not related to vowel height or sonority. It is also not related to tone, because the retracted forms can bear either tone, albeit this is dependent on the position of the vowel within a word. This interaction is phonetically unmotivated, and adds to the growing body of work on possible interactions between segmental and suprasegmental properties.

Our final contribution is methodological. Tonal wug-tests are rare and our work shows how they can be used to corroborate the data based on elicitation. In the paper, we outline a variety of challenges in recording the stimuli and interpreting the results. For instance, we prime the tonal stimuli with real words with the same prosodic patterns. Due to the gaps in the possible words, our statistical analysis is conducted on subsets of data rather than one single model across all data.

Data

The code for the analysis, experimental interface, and experimental data are available at: <https://doi.org/10.17605/OSF.IO/2ZSE9>.

CRedit authorship contribution statement

Gašper Beguš: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Peter Jurgec:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Tables 3–5

Table 3

Token counts for the elicitation data we analyzed acoustically. In Fig. 1, the data for long vowels include both Long Falling and Long Rising pitch. The numbers in brackets represent tokens with [ε] that are presented in Fig. 10.

	No clitic				With clitic (σ_{cl})			
	σ	$\sigma\sigma$			σ	$\sigma\sigma$		
		$\sigma_{\mu\mu}\sigma$	$\sigma_{\mu}\sigma$	$\sigma'\sigma$		$\sigma_{cl}\sigma$	$\sigma_{cl}\sigma\sigma$	$\sigma_{cl}\sigma\sigma\sigma$
Long Falling	93	102	102	47	34	68	39	21
Long Rising	57	68	46	8	17	31	6 (30)	19
Short	53	–	38	–	20	23	28	–
Total	203	170	195	55	71	122	103	40
Figures	1–3		4, 5		6		7–10	

Table 4

Stimuli for acoustic analysis.

Transcription	Length (σ)	Stress	Pitch	Retraction	Long	Gloss
yréex	1	Initial	Falling	No	Initial	sin
púot	1	Initial	Falling	No	Initial	track, path
jeésk	1	Initial	Rising	No	Initial	tongue, language
kodsts	1	Initial	Rising	No	Initial	reaper
yrát	1	Initial	Falling	Yes	Initial	castle
líet	1	Initial	Falling	Yes	Initial	ice
méex	1	Initial	Falling	Yes	Initial	bellows
nóof	1	Initial	Falling	Yes	Initial	night
núos	1	Initial	Falling	Yes	Initial	nose
péef	1	Initial	Falling	Yes	Initial	furnace
wrát	1	Initial	Falling	Yes	Initial	neck
yúópts	1	Initial	Rising	Yes	Initial	muzzle
jaápk	1	Initial	Rising	Yes	Initial	apple
sén	1	Initial	Short	No	None	hay
xórpt	1	Initial	Short	No	None	back
yróp	1	Initial	Short	Yes	None	grave
túf	1	Initial	Short	Yes	None	light
'bréezo	2	Initial	Falling	No	Initial	birch tree
'kráwo	2	Initial	Falling	No	Initial	cow
'tséesto	2	Initial	Falling	No	Initial	road
'kaáso	2	Initial	Rising	No	Initial	scythe
'kaázo	2	Initial	Rising	No	Initial	goat
'sédló	2	Initial	Rising	No	Initial	saddle
'zémljō	2	Initial	Rising	No	Initial	earth, dirt
'brádo	2	Initial	Falling	Yes	Initial	beard, chin
'hádwo	2	Initial	Falling	Yes	Initial	head
'éézer	2	Initial	Rising	Yes	Initial	lake
'soóntse	2	Initial	Rising	Yes	Initial	sun
'wádyo	2	Initial	Rising	Yes	Initial	scale
'ótrák	2	Initial	Falling	No	Peninitial	child
'yótuop	2	Initial	Falling	No	Peninitial	pigeon
'kókóof	2	Initial	Falling	No	Peninitial	chicken

Table 4 (continued)

Transcription	Length (σ)	Stress	Pitch	Retraction	Long	Gloss
'kór'wáw	2	Initial	Falling	No	Peninitial	bloody-ADJ
'lésien	2	Initial	Falling	No	Peninitial	woody
'úéséew	2	Initial	Falling	No	Peninitial	happy
'uétjfer	2	Initial	Falling	No	Peninitial	evening
'ímie	2	Initial	Falling	No	Peninitial	name
'ótruók	2	Initial	Rising	No	Peninitial	child-DAT
'kókoóf	2	Initial	Rising	No	Peninitial	chicken-GEN
'kórwaáw	2	Initial	Rising	No	Peninitial	bloody-ADV
'lésien	2	Initial	Rising	No	Peninitial	woody-NEUTR
'pámtáát	2	Initial	Rising	No	Peninitial	in spring
'póziim	2	Initial	Rising	No	Peninitial	in winter
'snézeén	2	Initial	Rising	No	Peninitial	snowy
'zópiét	2	Initial	Rising	No	Peninitial	tied
dá'múu	2	Peninitial	Falling	No	Peninitial	home
ko'léen	2	Peninitial	Falling	No	Peninitial	knee
lo'dii	2	Peninitial	Falling	No	Peninitial	people-GEN
moɣ'léen	2	Peninitial	Falling	No	Peninitial	foggy
po'léen	2	Peninitial	Falling	No	Peninitial	log
zi'dáar	2	Peninitial	Falling	No	Peninitial	mason
ot'raáts	2	Peninitial	Rising	No	Peninitial	children
dá'jéétfk	2	Peninitial	Rising	No	Peninitial	baby
'kótu	2	Initial	Short	No	None	wheel
'lipo	2	Initial	Short	No	None	linden tree
'xíje	2	Initial	Short	No	None	house

Table 5

Mean proportions of retraction ("Mean retr.") by word in the experiment. "N" stands for the number of judgments across all participants. The items are ordered by word length, then tone/length, then V(owel), and finally, mean retraction.

Item	Word Length	Tone/Length	V	Mean Retr.	N	Item	Word Length	Tone/Length	V	Mean Retr.	N
'zaámo	disyllable	Rising	aa	.750	36	'páamo	disyllable	Falling	aa	.485	33
'taáno	disyllable	Rising	aa	.719	32	'záano	disyllable	Falling	aa	.484	31
'waápo	disyllable	Rising	aa	.677	31	'wáano	disyllable	Falling	aa	.467	30
'baámo	disyllable	Rising	aa	.657	35	'ráaxo	disyllable	Falling	aa	.455	33
'raáxo	disyllable	Rising	aa	.647	34	'tááto	disyllable	Falling	aa	.452	31
'taábo	disyllable	Rising	aa	.629	35	'táabo	disyllable	Falling	aa	.429	35
'raáro	disyllable	Rising	aa	.588	34	'ráapo	disyllable	Falling	aa	.389	36
'paámo	disyllable	Rising	aa	.576	33	'wáapo	disyllable	Falling	aa	.364	33
'taáto	disyllable	Rising	aa	.500	36	'záamo	disyllable	Falling	aa	.346	26
'raápo	disyllable	Rising	aa	.486	37	'báamo	disyllable	Falling	aa	.258	31
'raáyo	disyllable	Rising	aa	.485	33	'kéeto	disyllable	Falling	ee	.526	38
'taáxo	disyllable	Rising	aa	.469	32	'méepo	disyllable	Falling	ee	.500	34
'teéno	disyllable	Rising	ee	.758	33	'réebo	disyllable	Falling	ee	.500	34
'reébo	disyllable	Rising	ee	.710	31	'réeto	disyllable	Falling	ee	.500	32
'peémo	disyllable	Rising	ee	.677	31	'réexo	disyllable	Falling	ee	.429	35
'leébo	disyllable	Rising	ee	.667	33	'méeso	disyllable	Falling	ee	.400	30
'zeémo	disyllable	Rising	ee	.657	35	'téebo	disyllable	Falling	ee	.257	35
'reéto	disyllable	Rising	ee	.636	33	'tóoro	disyllable	Falling	oo	.639	36
'teéxo	disyllable	Rising	ee	.613	31	'tóomo	disyllable	Falling	oo	.581	31
'keéto	disyllable	Rising	ee	.567	30	'móopo	disyllable	Falling	oo	.563	32
'reéxo	disyllable	Rising	ee	.531	32	'móoso	disyllable	Falling	oo	.429	35
'teébo	disyllable	Rising	ee	.514	35	'móoro	disyllable	Falling	oo	.400	40
'ueéto	disyllable	Rising	ee	.455	33	'wáan	monosyllable	Falling	aa	.765	34
'beémo	disyllable	Rising	ee	.394	33	'záan	monosyllable	Falling	aa	.607	28
'peémo	disyllable	Rising	ee	.472	36	'páam	monosyllable	Falling	aa	.600	35
'reéto	disyllable	Rising	ee	.433	30	'xáat	monosyllable	Falling	aa	.588	34
'reéxo	disyllable	Rising	ee	.364	33	'méep	monosyllable	Falling	ee	.806	36
'teéxo	disyllable	Rising	ee	.344	32	'mées	monosyllable	Falling	ee	.667	36
'reébo	disyllable	Rising	ee	.323	31	'tóor	monosyllable	Falling	oo	.879	33
'leébo	disyllable	Rising	ee	.294	34	'tóom	monosyllable	Falling	oo	.833	36
'teébo	disyllable	Rising	ee	.273	33	'móop	monosyllable	Falling	oo	.647	34
'ueéto	disyllable	Rising	ee	.257	35	'móos	monosyllable	Falling	oo	.600	35
'teéno	disyllable	Rising	ee	.250	36	'móor	monosyllable	Falling	oo	.528	36
'beémo	disyllable	Rising	ee	.235	34	'jón	monosyllable	Short	o	.676	34
'keéto	disyllable	Rising	ee	.235	34	'póx	monosyllable	Short	o	.571	36
'zeémo	disyllable	Rising	ee	.171	35	'tóp	monosyllable	Short	o	.545	35
'ráayo	disyllable	Falling	aa	.576	33	'sók	monosyllable	Short	o	.429	35
'táaxo	disyllable	Falling	aa	.571	35	'tós	monosyllable	Short	e	.563	32
'ráaro	disyllable	Falling	aa	.556	27	'rél	monosyllable	Short	e	.556	36
'xáato	disyllable	Falling	aa	.517	29	'uix	monosyllable	Short	i	.611	33
'táano	disyllable	Falling	aa	.514	37	'rík	monosyllable	Short	i	.600	35

References

- Adelson-Goldstein, J., & Shapiro, N. (2016). *Oxford Picture Dictionary*. Oxford: Oxford University Press.
- Akinlabi, A. (1996). Featural affixation. *Journal of Linguistics*, 32, 239–289.
- Anderson, S. R. (2005). *Aspects of the Theory of Clitics*. Oxford: Oxford University Press.
- Anderson, S. R. (2011). Clitics. In M. van Oostendorp, C. J. Ewen, E. Hume, & K. D. Rice (Eds.), *The Blackwell Companion to Phonology* (pp. 2002–2018). Malden, MA: Blackwell.
- Becker, M., & Gouskova, M. (2016). Source-oriented generalizations as grammar inference in Russian vowel deletion. *Linguistic Inquiry*, 47, 391–425.
- Becker, M., & Jurgec, P. (2017). Interactions of tone and ATR in Slovenian. In W. Kehrein, B. Köhnlein, P. Boersma, & M. van Oostendorp (Eds.), *Segmental Structure and Tone* (pp. 11–26). Berlin: De Gruyter.
- Becker, M., & Jurgec, P. (2020). Positional faithfulness drives laxness alternations in Slovenian. *Phonology*, 37, 335–366.
- Becker, M., Ketrez, N., & Nevins, A. (2011). The surfeit of the stimulus: Analytic biases filter lexical statistics in Turkish devoicing neutralization. *Language*, 87, 84–125.
- Becker, M. & Levine, J. (2012). Experigen – An online experiment platform. Available at <https://github.com/tlozoot/experigen>.
- Beckers, G. J. L. (2002). *rms equalize*. Praat script.
- Beckman, J. N. (1997). Positional faithfulness, positional neutralization and Shona vowel harmony. *Phonology*, 14, 1–46.
- Beckman, M. E., & Edwards, J. (1990). Lengthening and shortening and the nature of prosodic constituency. In J. Kingston & M. E. Beckman (Eds.), *Papers in Laboratory Phonology I: Between the Grammar and Physics of Speech* (pp. 152–178). Cambridge: Cambridge University Press.
- Beguš, G., & Dąbkowski, M. (2025). The blurring history of intervocalic devoicing. *Journal of Linguistics*, 61, 211–230.
- Beguš, G. (2011). Relativna kronologija naglasnih pojavov govora Žirovske kotline poljanskega narečja. *Slovenski jezik / Slovene Linguistic Studies*, 8, 19–33.
- Beguš, G. (2017). Effects of ejective stops on preceding vowel duration. *The Journal of the Acoustical Society of America*, 142, 2168–2184.
- Beguš, G. (2018). *Unnatural phonology: A synchrony-diachrony interface approach*. Ph.D. dissertation. Cambridge, MA: Harvard University.
- Beguš, G. (2019). Post-nasal devoicing and the blurring process. *Journal of Linguistics*, 55, 689–753.
- Beguš, G. (2020). Estimating historical probabilities of natural and unnatural processes. *Phonology*, 37, 515–549.
- Beguš, G. (2022). Distinguishing cognitive from historical influences in phonology. *Language*, 98, 1–34.
- Bennett, R., & Henderson, R. (2013). Accent in Uspanteko. *Natural Language & Linguistic Theory*, 31, 589–645.
- Benua, L. (1997). *Transderivational identity: Phonological relations between words*. Ph.D. dissertation. Amherst: University of Massachusetts.
- Berko, J. (1958). The child's learning of English morphology. *Word*, 14, 150–177.
- Bethin, C. Y. (1998). *Slavic Prosody: Language Change and Phonological Theory*. Cambridge: Cambridge University Press.
- Bidwell, C. A. (1969). *Outline of Slovenian Morphology*. Pittsburgh, PA: University Center for International Studies, University of Pittsburgh.
- Blevins, J. (2004). *Evolutionary Phonology: The Emergence of Sound Patterns*. Cambridge: Cambridge University Press.
- Blumenfeld, L.A. (2006). *Constraints on phonological interactions*. Ph.D. dissertation, Stanford University, Palo Alto, CA.
- Boersma, P. & Weenink, D. (1992). Praat. <http://www.praat.org>.
- Bolker, B. (2022). GLMM FAQ. Retrieved June 28, 2023.
- Božič, J. (2015). *Spell-out of phonological domains: The case of Slovenian*. Master's thesis. Vancouver, BC: University of British Columbia.
- Bradshaw, M. M. (1997). A phonology-phonetics mismatch: [voice] in consonant-tone interaction. *Studies in Linguistic Sciences*, 27, 17–31.
- Bradshaw, M. M. (1999). *A crosslinguistic study of consonant-tone interactions*. Ph.D. dissertation. Columbus, OH: Ohio State University.
- Buckley, E. (2000). On the naturalness of unnatural rules. *Proceedings from the Second Workshop on American Indigenous Languages. UCSB Working Papers in Linguistics*, 9, 1–14.
- Burzio, L. (2005). Sources of paradigm uniformity. In L. J. Downing, T. A. Hall, & R. Raffelsiefen (Eds.), *Paradigms in Phonological Theory* (pp. 65–106). Oxford: Oxford University Press.
- Caballero, G., & Carroll, L. (2015). Tone and stress in Choguita Rarámuri (Tarahumara) word prosody. *International Journal of American Linguistics*, 81, 457–493.
- Cahill, M. (2001). The unusual tone system of Awad Bing. Unpublished manuscript. Paper presented at the 75th Annual Meeting of the Linguistic Society of America (Washington, January 4–7, 2001).
- Christdas, P. (1988). *The phonology and morphology of Tamil*. Ph.D. dissertation. Ithaca, NY: Cornell University.
- Coetzee, A. W., Beddor, P. S., Shedden, K., Styler, W., & Wissing, D. (2018). Plosive voicing in Afrikaans: Differential cue weighting and tonogenesis. *Journal of Phonetics*, 66, 185–216.
- Coetzee, A. W., & Pretorius, R. (2010). Phonetically grounded phonology and sound change: The case of Tswana labial plosives. *Journal of Phonetics*, 38, 404–421.
- Connell, B. (2002). Tone languages and the universality of intrinsic F0: Evidence from Africa. *Journal of Phonetics*, 30, 101–129.
- Crosswhite, K. (2001). *Vowel Reduction in Optimality Theory*. New York: Routledge.
- de Lacy, P. (2002). The interaction of tone and stress in Optimality Theory. *Phonology*, 19, 1–32.
- de Lacy, P. (2006). *Markedness: Reduction and Preservation in Phonology*. Cambridge: Cambridge University Press.
- de Lacy, P. (2007). The interaction of tone, sonority, and prosodic structure. In P. de Lacy (Ed.), *The Cambridge Handbook of Phonology* (pp. 281–307). Cambridge: Cambridge University Press.
- Dixon, R. M. W., & Aikhenvald, A. Y. (2002). *Word: A Cross-Linguistic Typology*. Cambridge: Cambridge University Press.
- Donnelly, S. S. (2007). *Aspects of tone and voice in Phuthi*. Ph.D. dissertation. Urbana-Champaign: University of Illinois.
- Dąbkowski, M., & Beguš, G. (2024). Complex diachronies of final nasalization in Austronesian and Dakota. *Glossa: A Journal of General Linguistics*, 9, 1–31.
- Ernestus, M., & Baayen, H. (2003). Predicting the unpredictable: Interpreting neutralized segments in Dutch. *Language*, 79, 5–38.
- Fourakis, M. (1991). Tempo, stress, and vowel reduction in American English. *The Journal of the Acoustical Society of America*, 90, 1816–1827.
- Fowler, C. A., & Brown, J. M. (1997). Intrinsic f₀ differences in spoken and sung vowels and their perception by listeners. *Perception and Psychophysics*, 59, 729–738.
- Franks, S., & King, T. H. (2000). *A Handbook of Slavic Clitics*. Oxford: Oxford University Press.
- Gao, J., & Kirby, J. (2024). Laryngeal contrast and sound change: The production and perception of plosive voicing and co-intrinsic pitch. *Language*, 100, 124–158.
- Goldsmith, J. A. (1976). *Autosegmental phonology*. Ph.D. dissertation. Cambridge: Massachusetts Institute of Technology.
- Gonzales-Gomez, N., Poltrock, S., & Nazzi, T. (2013). A “bat” is easier to learn than a “tab”: Effects of relative phonotactic frequency on infant word learning. *PLOS ONE*, 8, 1–11.
- Gouskova, M. (2010). The phonology of boundaries and secondary stress in Russian compounds. *The Linguistic Review*, 27, 387–448.
- Gouskova, M., & Becker, M. (2013). Noun words show that Russian yer alternations are governed by the grammar. *Natural Language & Linguistic Theory*, 31, 735–765.
- Gouskova, M., & Roon, K. (2009). Interface constraints and frequency in Russian compound stress. In M. Babyonyshev, D. Kavitskaya, & J. Reich (Eds.), *Formal Approaches to Slavic Linguistics 17: The Yale Meeting 2008* (pp. 49–63). Ann Arbor, MI: Michigan Slavic Publications.
- Greenberg, M. L. (1987). Prozodične možnosti v slovenskem knjižnem jeziku in slovenskih narečjih. *Slavistična Revija*, 35, 171–186.
- Greenberg, M. L. (2000). *A Historical Phonology of Slovene*. Heidelberg: C. Winter Universitätsverlag.
- Greenberg, M. L. (2003). Word prosody in Slovene from a typological perspective. *STUF - Language Typology and Universals*, 56, 234–251.
- Gregerson, K. J. (1976). Tongue-root and register in Mon-Khmer. In Jenner, P.N., Thompson, L.C. & Starosta, S. (Eds.) *Austroasiatic Studies, Vol 1* (pp. 323–369). Honolulu: University of Hawaii Press.
- Gussenhoven, C. (2004). *The Phonology of Tone and Intonation*. Cambridge: Cambridge University Press.
- Gvozdanović, J. (1980). *Tone and Accent in Standard Serbo-Croatian*. Wien: Österreichische Akademie der Wissenschaften.
- Hayes, B. (1995). *Metrical Stress Theory: Principles and Case Studies*. Chicago: University of Chicago Press.
- Hayes, B., Zuraw, K., Siptár, P., & Londe, Z. (2009). Natural and unnatural constraints in Hungarian vowel harmony. *Language*, 85, 822–863.
- Herrity, P. (2000). *Slovene: A Comprehensive Grammar*. London, New York: Routledge.
- Hombert, J.-M. (1978). Consonant types, vowel quality and tone. In V. A. Fromkin (Ed.), *Tone: A Linguistic Survey* (pp. 77–112). New York: Academic Press.
- Horst, J. S., & Hout, M. C. (2016). The novel object and unusual name (noun) database: A collection of novel images for use in experimental research. *Behaviour Research Methods*, 48, 1393–1409.
- Hyde, B. (2002). A restrictive theory of metrical stress. *Phonology*, 19, 313–359.
- Hyman, L. M. (2006). Word-prosodic typology. *Phonology*, 23, 225–257.
- Hyman, L. M. (2009). How (not) to do phonology typology: The case of pitch-accent. *Language Sciences*, 31, 213–238.
- Hyman, L. M., & Leben, W. R. (2021). Tone systems. In C. Gussenhoven & A. Chen (Eds.), *The Oxford Handbook of Language Prosody* (pp. 45–65). Oxford: Oxford University Press.
- Inkelas, S., & Zec, D. (1988). Serbo-Croatian pitch accent: The interaction of tone, stress, and intonation. *Language*, 64, 227–248.
- Ivić, P. (1958). *Die serbokroatischen Dialekte: Ihre Struktur und Entwicklung: Bd. 1. Allgemeines und die štokavische Dialektgruppe*. 's Gravenhage: Mouton.
- Jurjec, P. (2004). Fonološke značilnosti novejšega slovenskega besedja. In M. Stabej (Ed.) *Moderno v slovenskem jeziku, literaturi in kulturi: 42. seminar slovenskega jezika, literature in kulturi*, 28. 6.–16. 7. 2004, (pp. 179–181). Ljubljana: Center za slovenščino kot drugi/tuji jezik pri Oddelku za slovenistiko Filozofske fakultete.
- Jurjec, P. (2007). *Novejše besedje s stališča fonologije: Primer slovenščine*. Ph.D. dissertation. University of Ljubljana.
- Jurjec, P. (2010). O prihodnosti fonologije slovenščine in v Sloveniji. In V. Gorjanc & A. Žele (Eds.), *Izzivi sodobnega jezikoslovja* (pp. 13–34). Znanstvena založba Filozofske fakultete: Ljubljana.
- Jurjec, P. (2011). Slovenščina ima 9 samoglasnikov. *Slavistična Revija*, 59, 243–268.
- Jurjec, P. (2016). Velar palatalization in Slovenian: Local and long-distance interactions in a derived environment effect. *Glossa*, 1, 24.
- Jurjec, P. (2019). Opacity in Šmartno Slovenian. *Phonology*, 36, 265–301.
- Jurjec, P. (2023). Accent systems, suprasegmental phonetics, and phonology. In M. L. Greenberg & L. A. Grenoble (Eds.), *Encyclopedia of Slavic Languages and Linguistics Online*. Brill.
- Jurjec, P., & Schertz, J. (2020). Postalveolar co-occurrence restrictions in Slovenian. *Natural Language & Linguistic Theory*, 38, 499–537.

- Kager, R. (2007). Feet and metrical stress. In P. de Lacy (Ed.), *The Cambridge Handbook of Phonology* (pp. 195–227). Cambridge: Cambridge University Press.
- Kapović, M. (2008). O naglasku u staroštokavskom slavonskom dialektu. *Croatica at Slavic Iadertina*, 4, 115–147.
- Kapović, M. (2015). *Povijest hrvatske akcentuacije: Fonetika*. Zagreb: Matica Hrvatska.
- Kenstowicz, M. (1997). *Quality-sensitive stress*. *Rivista di Linguistica*, 9, 157–187.
- Kenstowicz, M. (2004). Paradigmatic uniformity and contrast. In L. J. Downing, T. A. Hall, & R. Raffelsiefen (Eds.), *Paradigms in Phonological Theory* (pp. 145–169). Oxford: Oxford University Press.
- Kingston, J. (2007). Segmental influences on F0: Controlled or automatic. In Gussenhoven, C. & Riad, T. (Eds.), *Tones and Tunes*, vol. 2 (pp. 171–210). Berlin: Mouton de Gruyter.
- Kiparsky, P. (2006). Amphichronic program vs. evolutionary phonology. *Theoretical Linguistics*, 32, 217–236.
- Kiparsky, P. (2008). Universals constrain change, change results in typological generalizations. In J. Good (Ed.), *Linguistic Universals and Linguistic Change* (pp. 23–53). Oxford: Oxford University Press.
- Klatt, D. H. (1975). Vowel lengthening is syntactically determined in connected discourse. *Journal of Phonetics*, 3, 129–140.
- Köhnlein, B. (2016). Contrastive foot structure in Franconian tone accent dialects. *Phonology*, 33, 87–123.
- Ladd, R., & Silverman, K. E. A. (1984). Vowel intrinsic pitch in connected speech. *Phonetica*, 41, 31–40.
- Langston, K. (1997). Pitch accent in Croatian and Serbian: Towards an autosegmental analysis. *Journal of Slavic Linguistics*, 5, 80–116.
- Langston, K. (2006). *Čakavian Prosody: The Accentual Patterns of the Čakavian Dialects of Croatia*. Bloomington, IN: Slavica Publishers.
- Lee, S. J. (2008). *Consonant-tone interaction in Optimality Theory*. Ph.D. dissertation, Rutgers University, New Brunswick, NJ.
- Lehiste, I. (1961). Some acoustic correlates of accent in Serbo-Croatian. *Phonetica*, 7, 114–147.
- Lehiste, I. (1970). *Suprasegmentals*. Cambridge, MA: MIT Press.
- Lehiste, I., & Ivić, P. (1963). *Accent in Serbo-Croatian: An Experimental Study*. Ann Arbor: University of Michigan.
- Lehiste, I., & Ivić, P. (1986). *Word and Sentence Prosody in Serbo-Croatian*. Cambridge, MA: MIT Press.
- Lenček, R. L. (1966). *The Verb Pattern of Contemporary Standard Slovene: With an Attempt at a Generative Description of the Slovene Verb by Horace G. Lunt*. Bibliotheca Slavica, Wiesbaden: Otto Harrassowitz.
- Łukaszewicz, B. (2018). Phonetic evidence for an interactive stress system: The issue of consonantal rhythm. *Phonology*, 35, 115–150.
- Łukaszewicz, B., & Molzcanow, J. (2018). Rhythmic stress in Ukrainian: Acoustic evidence of a bidirectional system. *Journal of Linguistics*, 54, 367–388.
- Moore-Cantwell, C. (2020). Weight and final vowels in the English stress system. *Phonology*, 37, 657–695.
- Myers, S. (1997). OCP effects in Optimality Theory. *Natural Language & Linguistic Theory*, 15, 847–892.
- Myrberg, S., & Riad, T. (2015). The prosodic hierarchy of Swedish. *Nordic Journal of Linguistics*, 38, 115–147.
- Nespor, M., & Vogel, I. (1986). *Prosodic Phonology*. Dordrecht: Foris.
- Newlin-Lukowicz, L. (2012). Polish stress: Looking for phonetic evidence of a bidirectional system. *Phonology*, 29, 271–329.
- Odden, D. (1988). Floating tones and contour tones in Kenyang. *Studies in African Linguistics*, 19, 1–34.
- Odden, D. (2011). Features impinging on tone. In J. A. Goldsmith, E. Hume, & L. Wetzels (Eds.), *Tones and Features: Phonetic and Phonological Perspectives* (pp. 81–107). Berlin: Walter de Gruyter.
- Ohala, J. J., & Eukel, B. W. (1987). Explaining the intrinsic pitch of vowels. In R. Channon & L. Shockey (Eds.), *In Honor of Ilse Lehiste* (pp. 207–215). Dordrecht: Foris.
- Oller, D. K. (1973). The effect of position in utterance on speech segment duration in English. *Journal of the Acoustical Society of America*, 54, 1235–1247.
- Pearce, M. (2007). *The interaction of tone with voicing and foot structure: Evidence from Kera phonetics and phonology*. Ph.D. dissertation. London: University of London.
- Pearce, M. (2009). Kera tone and voicing interactions. *Lingua*, 119, 846–864.
- Peperkamp, S. (1997). *Prosodic Words*. The Hague: Holland Academic Graphics.
- Piternann, M. (2000). Effect of speaking rate and contrastive stress on formant dynamics and vowel perception. *The Journal of the Acoustical Society of America*, 107, 3425–3437.
- Prince, A. (1990). Quantitative consequences of rhythmic organization. In Noske, M. & Deaton, K. (Eds.), *CLS: Parasession on the syllable in phonetics and phonology*, Chicago Linguistics Society, vol. 26. 355–398.
- R Core Team (2013). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.
- Ramovš, F. (1928). O premiku akcenta v tipih zvězdá, žená in maglá v slovenskem jeziku. *Lud Slovianški*, 1a, 48–61.
- Ramovš, F. (1936). *Kratka zgodovina slovenskega jezika 1*. Ljubljana: Akademski založba.
- Ramovš, F. (1950). Relativna kronologija slovenskih akcentskih pojavov. *Slavistična Revija*, 3, 16–23.
- Rasin, E. (2017). *The stress-encapsulation universal and phonological modularity*. Massachusetts Institute of Technology. Unpublished manuscript.
- Remijsen, B. (2002). Lexically contrastive stress accent and lexical tone in Ma'ya. In C. Gussenhoven & N. Warner (Eds.), *Laboratory Phonology 7* (pp. 585–614). Berlin and New York: Mouton de Gruyter.
- Riad, T. (2012). Culminativity, stress and tone accent in Central Swedish. *Lingua*, 122, 1352–1379.
- Rigler, J. (1970). Akcentske variante I. *Slavistična revija*, 28, 4–15.
- Ronelle, A. (2019). Tracking new elements in Bulgarian dialects. In J. J. Pennington, V. A. Friedman, & L. A. Grenoble (Eds.), *And Thus You Are Everywhere Honored: Studies Dedicated to Brian D. Joseph* (pp. 21–28). Bloomington, IN: Slavica Publishers.
- Schuh, R. G. (1971). Verb forms and verb aspects in Ngizim. *Journal of African Languages*, 10, 47–60.
- Shih, S.-H. (2016). Sonority-driven stress does not exist. In G. Hansson, A. Farris-Trimble, K. McMullin, & D. Pulleyblank (Eds.), *Supplemental Proceedings of the 2015 Annual Meeting on Phonology* (pp. 1–11). Washington, DC: Linguistic Society of America.
- Shih, S.-H. (2018). On the existence of sonority-driven stress in Gujarati. *Phonology*, 35, 327–364.
- Shih, S.-H., & de Lacy, P. (2020). Evidence for sonority-driven stress. *Catalan Journal of Linguistics*, 18, 9–40.
- Škofic, J. & Kenda Jež, K. (2021). Tonemska in kolikostna nasprotja v slovenskih narečjih. In T. Mirtič & M. Snoj (Eds.), *1. slovenski pravorični posvet. Razprave II. razreda*, 25 (pp. 192–216). Ljubljana: Slovenska akademija znanosti in umetnosti.
- Stang, C. S. (1957). *Slavic Accentuation*. Oslo: Universitetsforlaget.
- Stanonik, M. (1977). Govor Žirovske kotline in njenega obdobja. *Slavistična Revija*, 25, 293–309.
- Statistical Office of the Republic of Slovenia (2024). SI-STAT data portal demographics.
- Steriade, D. (1997). Lexical conservatism. In *Linguistics in the Morning Calm, Selected Papers from SICOL 1997* (pp. 157–179). Seoul: Linguistic Society of Korea, Hanshin Publishing House.
- Sussex, R., & Cuddebeck, P. (2006). *The Slavic Languages*. Cambridge: Cambridge University Press.
- Tang, K. (2008). *The phonology and phonetics of consonant-tone interaction*. Ph.D. dissertation, University of California, Los Angeles.
- Toporišič, J. (1976/2000). Slovenska slovnica. Maribor: Obzorja.
- van der Hulst, H. (2011). Pitch accent systems. In M. van Oostendorp, C. J. Ewen, E. Hume, & K. D. Rice (Eds.), *The Blackwell Companion to Phonology* (pp. 1003–1026). Malden, MA: Blackwell.
- Whalen, D. H., & Levitt, A. G. (1995). The universality of intrinsic F₀ of vowels. *Journal of Phonetics*, 23, 349–366.
- Zec, D. (1999). Footed tones and tonal feet: Rhythmic consistency in a pitch-accent language. *Phonology*, 16, 225–264.
- Zec, D., & Zsiga, E. C. (2022). Tone and stress as agents of cross-dialectal variation: The case of Serbian. In H. Kubozono, J. Ito, & A. Mester (Eds.), *Prosody and Prosodic Interfaces* (pp. 63–94). Oxford: Oxford University Press.
- Zhang, J., & Lai, Y. (2010). Testing the role of phonetic knowledge in Mandarin tone sandhi. *Phonology*, 27, 153–201.
- Zhang, J., Lai, Y., & Turnbull-Sailor, C. (2006). Wug-testing the 'tone circle' in Taiwanese. *WCCFL*, 25, 453–461.
- Zhang, J., & Liu, J. (2016). The productivity of variable disyllabic tone sandhi in Tianjin Chinese. *Journal of East Asian Linguistics*, 25, 1–35.
- Zsiga, E. C., & Zec, D. (2013). Contextual evidence for the representation of pitch accents in Standard Serbian. *Language and Speech*, 56, 69–104.
- Zwicky, A. M., & Pullum, G. K. (1983). Cliticization vs. inflection: English n't. *Language*, 59, 502–513.